Multi-Axis Modal-Reduced Physics-Informed Neural Networks for Structural Parameter Identification

A Generalised-Force Projection and Multi-Sensor Reference-Coordinate Framework for Stiffness and Damping Recovery in a Three-Dimensional MDOF Structure

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Abstract

Reduced-order inverse identification is the operation by which a distributed structural system, represented numerically by tens of thousands of degrees of freedom, is condensed into a small number of interpretable physical parameters — stiffness, damping, effective mass — that an engineer can compare against a design value, track over a service life, or update after a damage event. The condensation is rarely the difficulty. The difficulty is *consistency*: a reduced parameter is only meaningful when the observation, the mass, the stiffness and the applied force are all expressed in the same coordinate. An identification that fits a displacement record to machine precision while silently mixing coordinate conventions returns a number that is numerically exact and physically meaningless.

This chapter develops a physics-informed neural network (PINN) formulation in which coordinate consistency is enforced before learning begins, and demonstrates it on all three spatial axes of a three-dimensional finite element structure. The work proceeds from a mass-normalised modal basis of a steel support table (five bodies, 828 SOLID186 elements, 14,859 solver degrees of freedom, total structural mass 78,500 kg) solved in ANSYS Mechanical 2025 R2. Eight slab-only observation nodes are instrumented. Four of them are fused by least squares into a single modal coordinate and mapped onto one common physical reference coordinate — sensor S_C07, node 3461 — while a fifth node, S_C06, is withheld entirely from the fusion and used as an unseen *spatial* validation point. The applied load is not taken as the raw Workbench face force. Instead the actual finite element load vector, assembled from a 1 Pa surface pressure over the 20 m^2 loaded face, is projected onto the retained mode, $Q_r(t) = \boldsymbol{\phi}_r^{\mathsf{T}} \mathbf{f}_{\mathrm{FE}}(t)$, and divided by the reference modal amplitude to obtain the equivalent reference-coordinate force $F_{\mathrm{eq}}(t) = Q_r(t)/\phi_{\mathrm{ref}}$. In the vertical direction, where the slab response carries a non-negligible second contribution, an overdetermined four-sensor spatial system is solved for the two participating modal coordinates and only the Mode-4 component is retained as the learning target.

The learning machine itself remains deliberately scalar. A fully connected network of four hidden layers of width 64, preceded by a 128-frequency Fourier feature encoding and using sinusoidal activations, represents the reference displacement $u(t)$ as a continuous and infinitely differentiable surrogate. Velocity and acceleration are obtained by exact automatic differentiation of that surrogate — never by finite differencing of sampled data — and the reduced equation of motion $m^*\ddot{u} + C\dot{u} + Ku = F_{\mathrm{eq}}(t)$ is imposed as a residual penalty at 10,000 collocation points. Stiffness and damping enter as trainable scalars under a softplus reparameterisation that enforces positivity by construction. Only displacement is supplied as data, and 20 % of the record is withheld from the data term entirely.

The identification succeeded uniformly. Against the authoritative ANSYS modal reference at the common S_C07 coordinate, the recovered stiffness agreed to $-\mathbf{0.175}\,\%$ in X, $-\mathbf{0.207}\,\%$ in Y and $-\mathbf{0.081}\,\%$ in Z; the corresponding natural frequencies agreed to $-0.088\,\%$, $-0.104\,\%$ and $-0.041\,\%$. The displacement surrogate generalised to 2,000 held-out time samples never presented to the network, reaching coefficients of determination of 0.999993, 0.999987 and 0.999994 at relative root-mean-square errors of $0.264\,\%$, $0.355\,\%$ and $0.248\,\%$. The spatial preprocessing was independently corroborated: reconstruction at the withheld sensor S_C06 carried errors of $0.0097\,\%$ (X), $0.1559\,\%$ (Y) and $2.99 \times 10^{-9}\,\%$ (Z), and the vertical modal separation was confirmed by an entirely independent route — a modal-force ratio of $Q_7/Q_4 \approx 0.2125$ combined with dynamic compliance at the drive frequency predicts a root-mean-square coordinate ratio of $4.019\,\%$, against a measured $4.041\,\%$ from the displacement decomposition, a relative discrepancy of $0.55\,\%$. The identified damping coefficients, recovered without prior estimate in every direction, correspond to effective reduced-model damping ratios of $0.0238\,\%$, $0.0242\,\%$ and $0.0166\,\%$, and are interpreted throughout as effective reduced-model quantities rather than as material damping, consistent with the zero physical damping and 0.005 numerical damping of the source transient solution.

Two qualifications are stated here rather than buried. First, the network is not identifying a multi-degree-of-freedom system: by the time it is initialised, the finite element solve, the modal basis, the least-squares sensor fusion and the load projection have reduced the problem to a single-degree-of-freedom ordinary differential equation with known mass, known forcing and exactly two unknown scalars. Ordinary least squares solves that same estimation problem in 0.18–0.22 ms against 223–367 s for the network, a factor of order 10^6, and recovers the same stiffness to within 0.009–0.041 %. The network is therefore defended on one ground only — it never differentiates the sampled record, and so escapes the noise amplification that drives the least-squares estimator to physically impossible negative damping in all three directions. Second, the vertical stiffness result must not be read as evidence that the two-mode decomposition produced superior accuracy: the vertical non-dimensionalisation places the reference at $\hat{k}_{\mathrm{ref}} = 1.0095$ against a global initialisation of $\hat{k}_0 = 1.0$, so that run began $0.94\,\%$ from its target and barely traversed parameter space. Its vertical damping coefficient, correspondingly, fluctuated by $16.95\,\%$ over the final tenth of iterations.

The defensible methodological claim is therefore narrower than uniformity of accuracy: it is that coordinate consistency, constructed before learning, is what makes the reduced inverse problem well posed in each direction, and that the same network, loss, optimiser and configuration path sufficed for all three with no architectural modification.

Keywords: physics-informed neural networks, inverse structural identification, modal reduction, generalised force projection, multi-sensor modal fusion, Fourier feature encoding, automatic differentiation, reduced-order model consistency, DeepXDE.

Contents

Abstract **i**

1 Introduction **1**

1.1 Motivation: Parameter Identification for Three-Dimensional Structural Systems . 1

1.2 The Coordinate-Consistency Problem in Reduced-Order Identification 2

1.3 Phase-2 Formulation: Multi-Sensor Fusion and Generalised-Force Projection . . . 2

1.4 The Division of Labour: What the Network Is Asked to Estimate 4

1.5 Objectives and Contributions 5

1.6 Organisation of the Chapter 6

2 Literature Review **7**

2.1 Physics-Informed Machine Learning: Origins and Formulation 7

2.2 Inverse Identification of Structural Parameters with PINNs 7

2.3 Modal Reduction and the Consistency of Reduced-Order Inverse Models 8

2.4 Spatial Observability, Sensor Placement and Modal Filtering 8

2.5 Spectral Bias and the Fourier Feature Remedy 9

2.6 Classical Baselines: Regression on Differentiated Response 9

2.7 Identified Research Gap 10

3 Theoretical Foundations **11**

3.1 From the Continuum to the Semi-Discrete Equations of Motion 11

3.2 Mass-Normalised Modal Decomposition 12

3.3 Reference-Coordinate Reduction: Equivalent Mass and Stiffness 13

3.4 Generalised Force Projection and the Consistent Right-Hand Side 14

3.5 Two-Mode Spatial Decomposition and Modal Filtering 15

3.6 Spectral Separation and Dynamic Magnification 16

3.7 Observability of Stiffness and Damping 18

3.8 The Neural Surrogate: Approximation and Exact Differentiation 19

3.9 Spectral Bias and Fourier Feature Encoding 19
3.10 The Composite Physics-Informed Loss Functional 20
3.11 Non-Dimensionalisation and Parameter Scaling 22
3.12 Positivity Enforcement by Softplus Reparameterisation 22
3.13 Joint Optimisation by Adam 23

4 Methodology **24**
4.1 Computational Environment and Execution 24
4.2 Software Architecture and the Direction-Agnostic Configuration Path 25
4.3 Finite Element Source: Modal and Transient Analyses 25
4.4 Observation Node Selection and Roles 26
4.5 Multi-Sensor Reference-Coordinate Fusion (X and Y) 26
4.6 Two-Mode Spatial Decomposition (Z) 27
4.7 Hidden-Sensor Spatial Validation 27
4.8 Exact Load-Vector Projection and the Equivalent Force 28
4.9 Dataset Freezing, Quality Control and Provenance 29
4.10 Network Construction and Fourier Encoding as Implemented 29
4.11 Collocation, Holdout and the Displacement-Only Data Term 29
4.12 Non-Dimensionalisation, Positivity and Optimiser Configuration 30
4.13 The Classical Least-Squares Diagnostic 30
4.14 Reproducibility Controls 31

5 Results and Discussion **32**
5.1 Overview 32
5.2 Preprocessing Fidelity: Spatial Validation at the Withheld Sensor 33
5.3 Independent Verification of the Vertical Modal Decomposition 34
5.4 Surrogate Accuracy and Temporal Generalisation 34
5.5 Identified Parameters 35
5.6 Agreement with the Authoritative Modal Reference 37
5.7 Agreement with the Classical Least-Squares Diagnostic 38
5.8 Parameter Convergence and Validated Stopping 42
5.9 Energy Audit and the Cost of Numerical Differentiation 44
5.10 Modal Purity as the Explanatory Variable 45
5.11 Evidence Against Memorisation 47
5.12 Effective Damping and its Interpretation 48

5.13 Limitations 49
5.14 Synthesis 49

6 Conclusion and Future Work **51**
6.1 Summary of Contributions 51
6.2 Future Work 52
6.3 Closing Remarks 53

References **55**

List of Figures

1.1 Phase-2 slab-only observation-node selection 3
1.2 The Phase-2 coordinate-consistency reduction chain 4

2.1 Positioning of the present work against the reviewed literature 10

3.1 Reference-coordinate displacement spectrum, X direction 17
3.2 Dynamic compliance of the reference-coordinate oscillators 17
3.3 Forward architecture of the identification network 19
3.4 Fourier feature bands against extracted modal content 20
3.5 Computational graph of the composite objective 21

4.1 End-to-end Phase-2 identification pipeline 25
4.2 Reference-coordinate displacement spectrum, Z direction 28

5.1 Predicted against reference displacement, X direction 36
5.2 Predicted against reference displacement, Y direction 36
5.3 Predicted against reference displacement, Z direction 37
5.4 Identified stiffness by method, X direction 38
5.5 Identified stiffness by method, Y direction 39
5.6 Identified stiffness by method, Z direction 39
5.7 Uniformity of identification accuracy across the three axes 40
5.8 Parameter convergence, X direction 43
5.9 Parameter convergence, Y direction 44
5.10 Parameter convergence, Z direction 44
5.11 Modal purity against identification accuracy 47

List of Tables

3.1 Finite element model summary. 12
3.2 Extracted modal frequencies and directional character. 13
3.3 Reference-coordinate equivalents at S_C07 (node 3461). 14
3.4 Exact finite element load-vector projection (180 loaded elements, $A = 20.000\,\mathrm{m}^2$, $p_{\max} = 555.15\,\mathrm{Pa}$). 15
3.5 Observability diagnostics over the 9.999 s record. 18
3.6 Scaling constants and resulting reference magnitudes. 22

4.1 Production run configuration and computational cost. 24
4.2 Observation nodes and final roles. 26
4.3 Mass-normalised modal amplitudes at the fusion sensors. 27
4.4 Network and encoding configuration. 30

5.1 Consolidated Phase-2 identification result, reference coordinate S_C07 / node 3461. 32
5.2 Preprocessing fidelity and spatial validation at the withheld sensor S_C06. . . . 33
5.3 Surrogate accuracy and temporal holdout validation. 35
5.4 Identified parameters, scaled and physical. 35
5.5 Agreement with the ANSYS modal reference at S_C07. 38
5.6 Classical least-squares diagnostic. 40
5.7 Computational cost of the two estimators on the identical frozen datasets. The least-squares timing is the mean of 200 repeated solves of the rank-2 normal equations over all 10,000 samples on a single CPU core; the network timing is the recorded wall-clock of the validated production run on a cluster GPU. The comparison is of end-to-end cost to obtain (K, C), not of like-for-like hardware. . 41
5.8 Parameter convergence and stability. 42
5.9 Energy audit of the identified models (finite-difference derivatives). 45
5.10 Local modal purity at S_C07 against downstream accuracy. 46

Chapter 1

Introduction

1.1 Motivation: Parameter Identification for Three-Dimensional Structural Systems

The structural engineer who wishes to know whether a built structure still behaves as designed has, in principle, a direct route available: excite it, measure its response, and infer the physical properties that produced that response. This inverse route underpins structural health monitoring, finite element model updating, digital-twin calibration, and post-event condition assessment. Its appeal is that it interrogates the structure as it actually is, rather than as its drawings say it should be.

The route is, however, considerably narrower in practice than in principle, and the narrowing occurs at a specific point. A real structure is a three-dimensional continuum. Its finite element representation carries thousands to millions of degrees of freedom. The quantity an engineer wants — "the lateral stiffness of this frame", "the damping available in the first vertical mode" — is a low-dimensional summary of that high-dimensional object, and the mapping from the object to the summary is not unique. It depends on where the structure was observed, which of its modes participated in the observed motion, and how the applied load distributed itself across those modes. Two identification exercises on the same structure, differing only in the node selected for observation, can return stiffness values differing by an order of magnitude, and both can be internally self-consistent.

This is not a defect of any particular estimator. It is a property of reduction. The reduced parameter is defined *relative to a coordinate*, and until that coordinate is fixed and the mass, stiffness, observation and force are all referred to it, the parameter has no determinate value. The present work takes this observation as its organising principle.

1.2 The Coordinate-Consistency Problem in Reduced-Order Identification

Consider a structure whose semi-discrete equations of motion are

$$\mathbf{M}\ddot{\mathbf{u}}(t) + \mathbf{C}\dot{\mathbf{u}}(t) + \mathbf{K}\mathbf{u}(t) = \mathbf{f}(t), \qquad \mathbf{u}(t) \in \mathbb{R}^n,$$

with $n = 14{,}859$ for the model examined here. A single-degree-of-freedom identification asserts the existence of scalars m^*, C, K and a scalar forcing $F(t)$ such that a scalar observation $u(t)$ obeys

$$m^*\ddot{u}(t) + C\dot{u}(t) + Ku(t) = F(t).$$

Four separate consistency requirements are hidden in that assertion, and each can fail independently.

1. Observation consistency. The scalar $u(t)$ must be the response of *one* mode. A raw nodal displacement is a superposition $u_j(t) = \sum_r \phi_{r,j} q_r(t)$ over every participating mode. Fitting a single-oscillator equation to a multi-mode superposition produces a stiffness that is a frequency-weighted average of several modal stiffnesses, belonging to none of them.

2. Mass consistency. The scalar m^* is not nodal mass and is not total structural mass. Under mass normalisation it is $m_j^* = \phi_{r,j}^{-2}$, an amplitude-dependent quantity that changes by orders of magnitude between observation points on the same structure.

3. Force consistency. The scalar $F(t)$ is not the resultant applied load. It is the modal participation of that load, $Q_r(t) = \boldsymbol{\phi}_r^{\mathsf{T}}\mathbf{f}(t)$, referred back to the observation coordinate. A distributed load that is large in magnitude but poorly aligned with the retained mode shape contributes little to Q_r, and using its resultant in place of Q_r/ϕ_{ref} inflates the identified stiffness by exactly the misalignment factor.

4. Spatial support. A single observation point provides no means of detecting when requirements 1–3 have been violated. If the selected node happens to sit near a nodal line of the intended mode, $\phi_{r,j} \to 0$, the equivalent mass diverges, and the identification silently transfers to whichever mode happens to dominate locally.

These four requirements are independent of the estimator. They bind least-squares regression, subspace identification, frequency-domain modal fitting, and physics-informed neural networks alike. The contribution of a learning method, if any, must therefore lie elsewhere — and it does, as Section 1.3 sets out.

1.3 Phase-2 Formulation: Multi-Sensor Fusion and Generalised-Force Projection

The formulation developed here addresses each requirement above *before* any learning takes place, so that what the network is asked to identify is already a well-posed scalar problem.

Observation consistency is obtained by multi-sensor modal fusion. Eight nodes are instru-

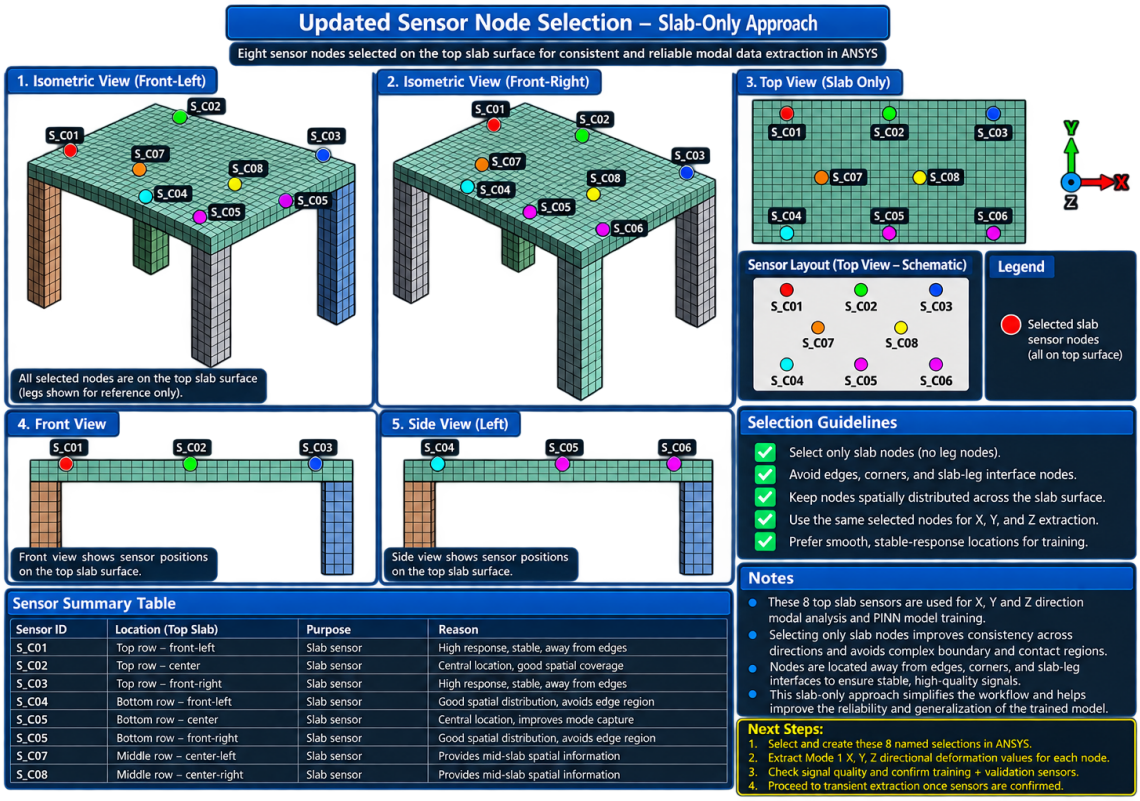


Figure 1.1: Phase-2 slab-only observation-node selection on the ANSYS model. Eight nodes are placed on the top slab surface and used identically across all three excitation directions. Four (S_C02, S_C05, S_C07, S_C08) are fused into the reference modal coordinate; S_C06 is withheld for spatial validation; the remainder are excluded from the final fusion.

mented on the top slab (Figure 1.1). Four are retained for fusion. For the two lateral directions, where a single mode dominates, the four records are mapped to a common modal coordinate and reconstructed at the reference node. For the vertical direction, where the slab motion contains a second participating mode, the four records form an overdetermined spatial system

$$u_j(t) = \phi_{4,j}\, q_4(t) + \phi_{7,j}\, q_7(t), \qquad j \in \{\mathrm{C02}, \mathrm{C05}, \mathrm{C07}, \mathrm{C08}\},$$

which is solved for both modal coordinates; only $u_{\mathrm{ref},4}(t) = \phi_{4,\mathrm{ref}}\, q_4(t)$ is retained as the learning target. The learning target in the vertical direction is therefore an isolated modal response, not a raw nodal displacement.

Mass consistency is obtained by fixing one physical reference coordinate — sensor S_C07 at node 3461 — for all three directions, and computing $m^* = \phi_{\mathrm{ref}}^{-2}$ from the mass-normalised mode shape at that node.

Force consistency is obtained by exact finite element load-vector projection. A unit surface pressure is assembled over the loaded face, projected onto each retained mode through the solver's own node-ordering map, and scaled by the physical pressure history to yield $Q_r(t)$; division by ϕ_{ref} returns the equivalent reference-coordinate force.

Spatial support is obtained by deliberately withholding one instrumented node, S_C06, from the fusion. The modal coordinate estimated from the four retained sensors is used to predict the

response at S_C06, and the prediction is compared against the finite element record there. This is a purely spatial test and is conceptually distinct from the temporal train/holdout split used later during network training; the two are never conflated.

Only after these four steps are complete is the reduced problem handed to the network. Figure 1.2 renders the resulting chain.

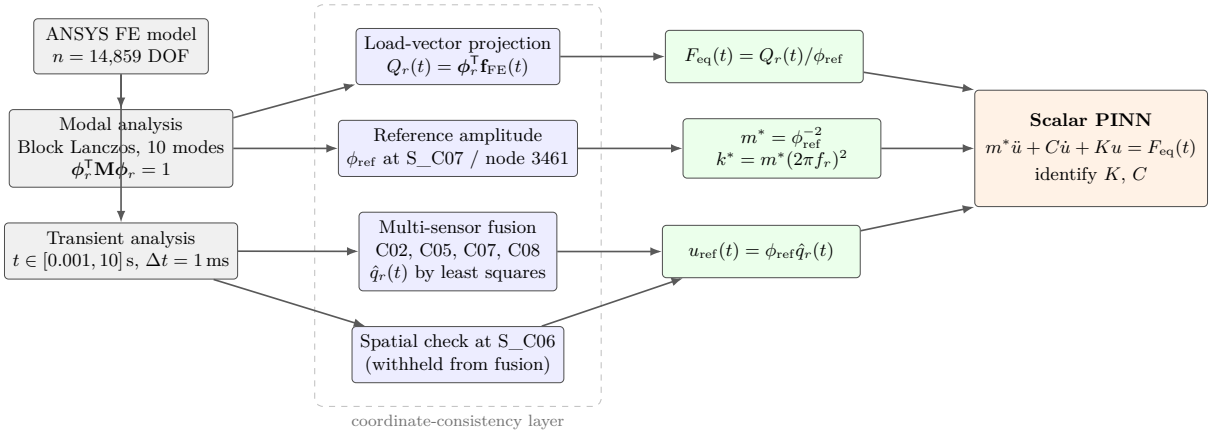


Figure 1.2: The Phase-2 reduction chain from the three-dimensional finite element model to the scalar physics-informed learning problem. Every quantity entering the reduced equation is referred to the single physical coordinate S_C07 before training begins.

1.4 The Division of Labour: What the Network Is Asked to Estimate

It is necessary to state at the outset, and precisely, how little of the physics in this chapter the neural network is responsible for. A claim that a physics-informed network "solves" the dynamics of a three-dimensional structure would be false here, and the construction of Section 1.3 is exactly what makes it false. The consistency layer is not a preliminary to the identification; it is the overwhelming majority of it.

By the time the network is initialised, the following have already been established by entirely classical means:

1. ANSYS has solved the full 14,859-degree-of-freedom elastodynamic problem, producing the transient response that the whole exercise takes as ground truth;
2. block Lanczos iteration has supplied the mass-normalised modal basis, from which the reference amplitude ϕ_{ref} fixes both the equivalent mass m^* *and* the reference stiffness k^* — that is, the quantity the network will later be credited with identifying is already known, exactly, before training begins;
3. an ordinary least-squares fit over four sensors has collapsed the spatial displacement field onto a single scalar modal coordinate;
4. an exact finite element load-vector projection has supplied the scalar forcing history.

What remains is one ordinary differential equation,

$$m^* \ddot{u}(t) + C \dot{u}(t) + K u(t) = F_{\mathrm{eq}}(t),$$

in which the mass is known, the forcing is known, the observation is known, and exactly two scalars are unknown. The network is a two-parameter estimator operating on a single-degree-of-freedom problem. It is not identifying a multi-degree-of-freedom system; the modal reduction did that, and the modal reduction is classical.

This framing matters because it sets the bar the network must clear. The same two-parameter problem is solved by ordinary least squares in 0.18–0.22 ms, against 223–367 s of GPU time for the network — a factor of order 10^6 — and the two agree on stiffness to within 0.009 %–0.041 % (Section 5.7). On speed the comparison is not close, and on stiffness accuracy there is nothing to choose between them. A method a million times more expensive and no more accurate requires a specific justification, and exactly one is available.

That justification is structural, not statistical. The least-squares estimator must be handed $\dot{u}$ and $\ddot{u}$. Obtaining them from a sampled record means differentiation, and differentiation amplifies high-frequency content by Δt^{-2}; obtaining them from solver exports means ingesting quantities that already carry the time-integration scheme's own numerical dissipation. Either route contaminates precisely the parameter — damping — that the dissipation term is meant to measure. The consequence is visible in this dataset: the unconstrained least-squares estimator returns *negative* damping, near -2.3 % of critical, in all three directions. The network never differentiates the data at all. Its velocity and acceleration are exact derivatives of a smooth surrogate constrained only by displacement, and its damping estimate is positive and physically admissible in every direction.

The case made in this chapter is therefore not that a PINN is a better estimator of K than ordinary least squares. It is that a derivative-free estimator remains admissible where a derivative-consuming one does not, and that the price of that admissibility — six orders of magnitude in computational cost — is worth stating plainly rather than leaving implicit.

1.5 Objectives and Contributions

The objectives of this chapter are as follows.

O1. To formulate a reduced-order inverse identification problem in which the observation, mass, stiffness and force are provably referred to one common physical coordinate, and to state the mathematical conditions under which the reduction is exact.

O2. To construct, from a mass-normalised modal basis and an exactly projected finite element load vector, a set of frozen learning datasets for the X, Y and Z directions of a three-dimensional structure.

O3. To identify stiffness and effective damping in each direction using a single scalar physics-informed network trained on displacement alone, with velocity and acceleration obtained exclusively by automatic differentiation.

O4. To validate the result along two independent axes: spatially, at an instrumented node withheld from the modal fusion; and temporally, at time samples withheld from the network's data term.

O5. To establish that the accuracy obtained is uniform across three structurally dissimilar directions, and to attribute that uniformity to the consistency construction rather than to network capacity.

The contributions are correspondingly fourfold.

C1 — Exact load-vector projection in place of force heuristics. The equivalent force is obtained from the assembled finite element load vector under the solver's own node ordering, not from a scalar conversion factor estimated from geometry. The resulting projection ratios differ materially between directions (1.0016, 1.0086 and 0.7685), and the vertical value in particular could not have been guessed.

C2 — Multi-sensor modal filtering as a preprocessing operation, not an architectural one. Spatial information from four sensors is used to isolate a single modal coordinate *before* training. The network remains a one-input, one-output scalar map. This keeps the inverse problem well-posed without enlarging the hypothesis space.

C3 — Dual, non-overlapping validation. Spatial generalisation is tested at a physically distinct, withheld finite element node; temporal generalisation is tested at withheld time samples. Neither test reuses information from the other.

C4 — Independent cross-verification of the modal decomposition. The vertical two-mode separation is confirmed by comparing the measured modal-coordinate amplitude ratio against a prediction formed from the projected modal forces and the analytical dynamic compliance at the drive frequency — two routes that share no intermediate quantity.

1.6 Organisation of the Chapter

Chapter 2 reviews the relevant literature across physics-informed machine learning, inverse structural identification, modal reduction, spatial observability and spectral bias, and states the research gap. Chapter 3 develops the theoretical foundations: the semi-discrete equations of motion, mass-normalised modal decomposition, reference-coordinate reduction, generalised force projection, two-mode spatial separation, observability of stiffness and damping, the neural surrogate and its spectral properties, and the composite loss functional. Chapter 4 describes the methodology as implemented. Chapter 5 presents and discusses the results for all three directions. Chapter 6 concludes.

Chapter 2

Literature Review

2.1 Physics-Informed Machine Learning: Origins and Formulation

The physics-informed neural network as it is now understood was consolidated by Raissi, Perdikaris and Karniadakis (2019), who showed that a differential operator can be evaluated on a network's output by automatic differentiation and penalised as a residual, so that a single optimisation simultaneously fits data and enforces a governing equation. The idea has antecedents — Dissanayake and Phan-Thien (1994) and Lagaris, Likas and Fotiadis (1998) both solved differential equations with neural approximants — but the modern form owes its practicality to reverse-mode automatic differentiation and to the framing of the inverse problem as joint optimisation over network weights and physical parameters.

The broader programme was set out by Karniadakis et al. (2021), who identify the inverse problem as the setting in which physics-informed learning is most clearly advantaged: where a forward solver competes against decades of optimised numerical analysis, an inverse solver competes against estimators that must first differentiate noisy sampled data. Baniya and Maity (2026) provide a recent and comprehensive survey specific to structural engineering, cataloguing architectural variants, training strategies and the distribution of applications across computational mechanics, damage detection and system identification.

2.2 Inverse Identification of Structural Parameters with PINNs

Application to structural parameter identification has developed quickly. Guo and Fang (2023) identified storey stiffnesses of five-storey steel frames by embedding regression-derived physical constraints in the loss, describing the result as a "grey-box" mechanism sitting between physical and data-driven models. Haghighat et al. (2021) established the corresponding formulation in solid mechanics, treating material parameters as trainable variables alongside network weights.

More recent work has extended the formulation in three directions relevant here. Teloli et al. (2025) developed a PINN framework for beam-like structures that poses the inverse problem in the frequency domain and identifies both elastic modulus — homogeneous and spatially varying

— and damping, with experimental validation on a cantilever steel beam. Liu et al. (2025) introduced a two-scale architecture that embeds small parameters directly in the network and reported parameter inversion errors below 10 % under noise for a T-shaped tower, validated both numerically and experimentally. Yamaguchi et al. (2024) applied a physics-informed formulation to nonlinear damage identification in a reinforced concrete bridge pier under seismic response.

Two observations follow from this body of work. First, the structural applications are dominated by systems whose reduction to a small parameter set is either trivial (a shear-building model with explicit storey stiffnesses) or analytically given (an Euler–Bernoulli beam). Second, where a full three-dimensional finite element model is the source, the question of which coordinate the identified parameter belongs to is seldom posed explicitly. Both observations motivate the present treatment.

2.3 Modal Reduction and the Consistency of Reduced-Order Inverse Models

The classical theory of modal reduction is long settled. Clough and Penzien (1993) and Chopra (2017) give the standard development: for a linear system with a symmetric positive-definite mass matrix and a stiffness matrix admitting the generalised eigenproblem $\mathbf{K}\boldsymbol{\phi}_r = \omega_r^2 \mathbf{M}\boldsymbol{\phi}_r$, the modal transformation decouples the equations of motion, and the generalised force on each mode is the projection of the physical load onto that mode's shape. Craig and Kurdila (2006) treat the corresponding component-mode and substructuring machinery.

What the classical theory supplies, and what is often left implicit in learning-based identification, is that the reduced stiffness is a property *of the pair* (mode, reference coordinate), not of the mode alone. Under the mass normalisation $\boldsymbol{\phi}_r^{\mathsf{T}} \mathbf{M} \boldsymbol{\phi}_r = 1$, the generalised stiffness is ω_r^2, dimensionally inconsistent with a physical stiffness; converting it to a physical K requires selecting a coordinate and dividing by the square of the mode shape amplitude there. Friswell and Mottershead (1995) make the analogous point in the model-updating literature, where the choice of reference degrees of freedom determines whether an updated parameter is physically interpretable.

2.4 Spatial Observability, Sensor Placement and Modal Filtering

The question of where to observe a structure so that a target mode is well conditioned is the subject of an extensive optimal-sensor-placement literature. Kammer (1991) introduced effective independence, which ranks candidate locations by their contribution to the determinant of the Fisher information matrix of the target mode set. Papadimitriou (2004) reformulated placement as information-entropy minimisation over the identified parameters. The shared conclusion is that a poorly placed sensor does not merely degrade an estimate; it can render a mode unobservable, at which point the identification converges confidently onto a different mode.

Modal filtering — the use of several simultaneous measurements to extract one modal coordinate — is treated in the experimental modal analysis literature by Ewins (2000) and Maia and Silva (1997). Its logic is straightforward: with s sensors and p participating modes, $s > p$ makes the

spatial system overdetermined and the modal coordinates recoverable by least squares, with the residual serving as a direct measure of whether the assumed modal content is complete. The present work applies exactly this construction, with $s = 4$ and $p \in \{1, 2\}$, and uses a fifth withheld sensor to test the result.

2.5 Spectral Bias and the Fourier Feature Remedy

Standard fully connected networks learn low-frequency components of a target function substantially faster than high-frequency components. Rahaman et al. (2019) established the phenomenon empirically as the "spectral bias"; Basri et al. (2019) and Cao et al. (2019) analysed it theoretically. The mechanism is made precise by the neural tangent kernel of Jacot, Gabriel and Hongler (2018): the training dynamics of a wide network are governed by a kernel whose eigenvalue spectrum decays with the frequency of the associated eigenfunction, so that high-frequency error components are suppressed at correspondingly slow rates.

For structural dynamics this is directly consequential. A ten-second record of a 51 Hz oscillation contains over five hundred cycles, and a plain coordinate network must represent all of them. Tancik et al. (2020) showed that mapping the input through a fixed set of sinusoidal features before the first layer flattens the kernel spectrum and restores the network's ability to fit high-frequency content. Wang, Wang and Perdikaris (2021) demonstrated the same remedy specifically within physics-informed training, where the residual term itself involves second derivatives and is therefore especially sensitive to the spectral deficiency. The architecture used in this chapter adopts a 128-frequency encoding spanning the band containing the relevant modal content.

2.6 Classical Baselines: Regression on Differentiated Response

The classical counterpart to the present method is linear regression on the equation of motion evaluated at sampled instants. Given displacement, velocity and acceleration histories, the equation $m^*\ddot{u} + C\dot{u} + Ku = F$ is linear in (K, C) and solvable by least squares. Juang and Pappa (1985) and Van Overschee and De Moor (1996) developed the realisation- and subspace-based alternatives that dominate operational modal analysis.

The structural weakness of the direct regression baseline is that it requires $\dot{u}$ and $\ddot{u}$ as inputs. When these are obtained by finite differencing a sampled record, differentiation amplifies high-frequency content and the estimator becomes sensitive to sampling rate and endpoint treatment. When, as here, they are available as solver exports, they are discrete solver quantities carrying the integration scheme's own numerical damping — which is precisely the quantity a damping identification is attempting to measure. In this chapter the least-squares estimator is therefore retained as a *diagnostic on the data*, not as a competing identification: it verifies that the frozen displacement, force and mass are mutually coherent, and its damping output is not interpreted as a physical damping estimate.

2.7 Identified Research Gap

Three gaps emerge from the foregoing.

Gap 1 — Coordinate consistency is assumed rather than constructed. Learning-based identification applied to three-dimensional finite element sources typically adopts a nodal response, a nominal applied load and a mass taken from the model, without establishing that the three belong to the same reduced coordinate. Where the reduction happens to be benign this is harmless; where it is not, the failure is silent.

Gap 2 — Force reduction is treated as a scaling detail. The distinction between the resultant applied load and its modal participation is well known classically but is rarely carried through into learning pipelines. The vertical direction of the present model exhibits a projection ratio of 0.7685 — a 23 % discrepancy that would propagate directly into the identified stiffness.

Gap 3 — Spatial and temporal validation are conflated. Held-out validation in PINN identification almost universally means held-out *time samples*. A record can be reproduced perfectly in time while the underlying spatial modal assumption is wrong. Spatial validation at a withheld physical location tests a different and complementary proposition.

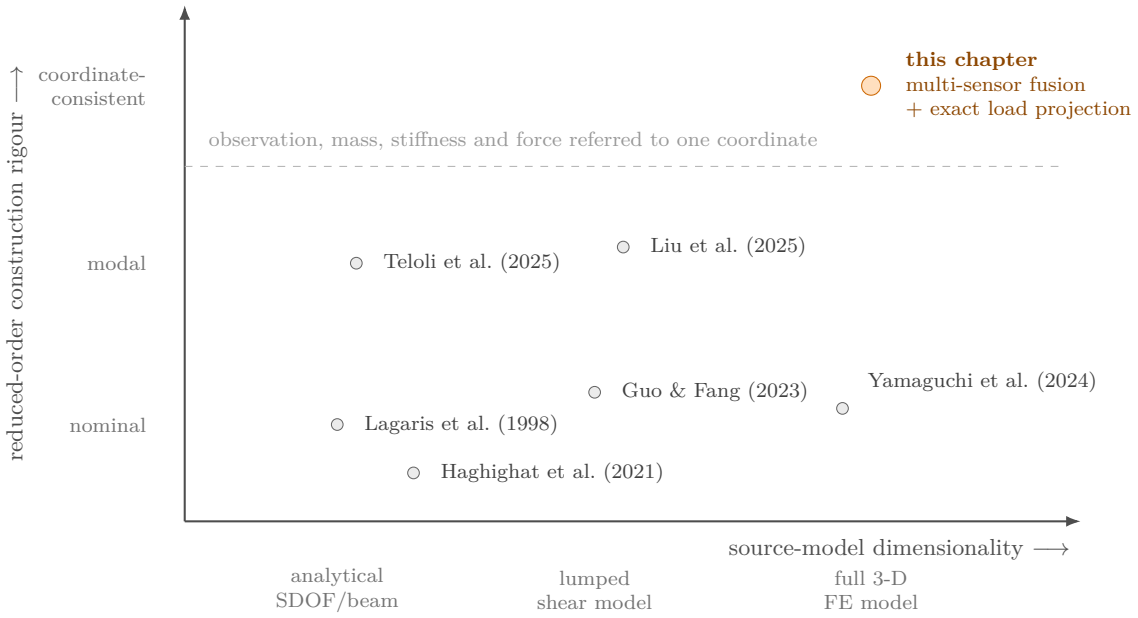


Figure 2.1: Positioning of the present work against the reviewed literature along two axes: the fidelity of the reduced-order construction, and the dimensionality of the source model.

Chapter 3

Theoretical Foundations

3.1 From the Continuum to the Semi-Discrete Equations of Motion

The structure is a linear elastic continuum occupying a region $\Omega \subset \mathbb{R}^3$ with displacement field $\mathbf{u}(\mathbf{x}, t)$. Momentum balance in the absence of body forces requires

$$\rho\, \ddot{\mathbf{u}}(\mathbf{x}, t) = \nabla \cdot \boldsymbol{\sigma}(\mathbf{x}, t), \qquad \mathbf{x} \in \Omega,$$

with the linear elastic constitutive relation $\boldsymbol{\sigma} = \mathbb{C} : \boldsymbol{\varepsilon}$, $\boldsymbol{\varepsilon} = \tfrac{1}{2}(\nabla \mathbf{u} + \nabla \mathbf{u}^{\mathsf{T}})$, and the isotropic tensor $\mathbb{C}$ determined by $E = 200\,\mathrm{GPa}$ and $\nu = 0.30$, with $\rho = 7850\,\mathrm{kg\,m^{-3}}$. The structure is fixed at the four leg bases, $\mathbf{u} = \mathbf{0}$ on Γ_D, and loaded by a prescribed traction $\bar{\mathbf{t}}(\mathbf{x}, t)$ on the loaded portion Γ_N of the top slab face.

The weak form follows from multiplying by an admissible virtual displacement $\mathbf{w}$ vanishing on Γ_D and integrating by parts:

$$\int_\Omega \rho\, \mathbf{w} \cdot \ddot{\mathbf{u}}\, \mathrm{d}\Omega + \int_\Omega \boldsymbol{\varepsilon}(\mathbf{w}) : \mathbb{C} : \boldsymbol{\varepsilon}(\mathbf{u})\, \mathrm{d}\Omega = \int_{\Gamma_N} \mathbf{w} \cdot \bar{\mathbf{t}}\, \mathrm{d}\Gamma.$$

Introducing the isoparametric discretisation $\mathbf{u}(\mathbf{x}, t) \approx \mathbf{N}(\mathbf{x})\mathbf{u}(t)$ over quadratic SOLID186 hexahedra and applying the same interpolation to $\mathbf{w}$ yields the semi-discrete system

$$\mathbf{M}\ddot{\mathbf{u}}(t) + \mathbf{C}\dot{\mathbf{u}}(t) + \mathbf{K}\mathbf{u}(t) = \mathbf{f}(t), \tag{3.1}$$

with

$$\mathbf{M} = \int_\Omega \rho\, \mathbf{N}^{\mathsf{T}}\mathbf{N}\, \mathrm{d}\Omega, \qquad \mathbf{K} = \int_\Omega \mathbf{B}^{\mathsf{T}}\mathbb{C}\,\mathbf{B}\, \mathrm{d}\Omega, \qquad \mathbf{f}(t) = \int_{\Gamma_N} \mathbf{N}^{\mathsf{T}}\bar{\mathbf{t}}(t)\, \mathrm{d}\Gamma,$$

where $\mathbf{B} = \partial\mathbf{N}$ is the strain–displacement operator. For the present model $\dim \mathbf{u} = 14{,}859$. The four bonded contacts between legs and slab are enforced by multipoint constraints with zero initial penetration; these eliminate degrees of freedom but do not alter the structure of (3.1). The physical damping matrix is $\mathbf{C} = \mathbf{0}$ in the source model, a point returned to in Section 3.7.

Table 3.1: Finite element model summary.

Quantity	Value
Leg cross-section	$0.5 \times 0.5\,\mathrm{m}$
Leg height	$3.0\,\mathrm{m}$
Top slab	$5 \times 4 \times 0.35\,\mathrm{m}$
Total structural volume	$10.0\,\mathrm{m}^3$
Young's modulus E	$200\,\mathrm{GPa}$
Poisson ratio ν	0.30
Density ρ	$7850\,\mathrm{kg\,m^{-3}}$
Mass per leg	$5{,}887.5\,\mathrm{kg}$
Top slab mass	$54{,}950\,\mathrm{kg}$
Total structural mass	$78{,}500\,\mathrm{kg}$
Nodes	$\approx 5{,}273$
SOLID186 elements	828
Contact elements	104
Solver degrees of freedom	14,859

3.2 Mass-Normalised Modal Decomposition

Setting $\mathbf{C} = \mathbf{0}$ and $\mathbf{f} = \mathbf{0}$ in (3.1) and seeking $\mathbf{u}(t) = \boldsymbol{\phi}\,\mathrm{e}^{\mathrm{i}\omega t}$ gives the generalised symmetric eigenproblem

$$\mathbf{K}\boldsymbol{\phi}_r = \omega_r^2\,\mathbf{M}\boldsymbol{\phi}_r, \qquad r = 1, \dots, n. \tag{3.2}$$

Because $\mathbf{M}$ is symmetric positive definite and $\mathbf{K}$ symmetric positive semi-definite, the eigenvalues ω_r^2 are real and non-negative and the eigenvectors are $\mathbf{M}$-orthogonal. The solver returns the first ten pairs by block Lanczos iteration, normalised such that

$$\boldsymbol{\phi}_r^{\mathsf{T}}\mathbf{M}\boldsymbol{\phi}_s = \delta_{rs}, \qquad \text{whence} \qquad \boldsymbol{\phi}_r^{\mathsf{T}}\mathbf{K}\boldsymbol{\phi}_s = \omega_r^2\,\delta_{rs}. \tag{3.3}$$

Expanding $\mathbf{u}(t) = \sum_r \boldsymbol{\phi}_r q_r(t)$ and premultiplying (3.1) by $\boldsymbol{\phi}_r^{\mathsf{T}}$ therefore gives n decoupled scalar equations

$$\ddot{q}_r(t) + 2\zeta_r\omega_r\dot{q}_r(t) + \omega_r^2 q_r(t) = Q_r(t), \qquad Q_r(t) \equiv \boldsymbol{\phi}_r^{\mathsf{T}}\mathbf{f}(t). \tag{3.4}$$

Equation (3.4) is the origin of everything that follows. Note what the normalisation has done: the modal mass is unity and the modal stiffness is ω_r^2, so neither carries the dimensions of a physical mass or stiffness. Recovering physically interpretable quantities requires the coordinate transformation of Section 3.3.

The two lateral modes are separated by only $0.57\,\mathrm{Hz}$, a spacing of $2.8\,\%$ — close enough that spatial rather than spectral discrimination is required, and a direct justification for the multi-sensor approach of Section 3.5.

Table 3.2: Extracted modal frequencies and directional character.

Mode	f_r (Hz)	Character	Effective modal mass (kg)	Participation
1	20.083511908708	dominant X bending	68,171.6	86.84 %
2	20.658102128190	dominant Y bending	68,417.3	87.16 %
3	30.515529745094	torsional	—	—
4	51.434203616507	principal Z slab	45,006.8	57.33 %
5	86.682696824013	higher slab	—	—
6	104.800744684277	higher slab	—	—
7	114.108085366480	secondary Z contribution	2,379	3.03 %
8	170.323346427785	higher order	—	—
9	173.255433148323	higher order	—	—
10	186.707792400038	higher order	—	—

3.3 Reference-Coordinate Reduction: Equivalent Mass and Stiffness

Let j denote a scalar physical degree of freedom — one node, one direction. Suppose the response at j is dominated by a single mode r, so that

$$u_j(t) \;\approx\; \phi_{r,j}\, q_r(t), \qquad \phi_{r,j} \equiv \mathbf{e}_j^{\mathsf{T}} \boldsymbol{\phi}_r. \tag{3.5}$$

Substituting $q_r = u_j/\phi_{r,j}$ into (3.4) and multiplying through by $\phi_{r,j}^{-1}$:

$$\frac{1}{\phi_{r,j}^2}\,\ddot{u}_j + \frac{2\zeta_r\omega_r}{\phi_{r,j}^2}\,\dot{u}_j + \frac{\omega_r^2}{\phi_{r,j}^2}\,u_j = \frac{Q_r(t)}{\phi_{r,j}}. \tag{3.6}$$

Equation (3.6) is a single-degree-of-freedom equation in the *physical* variable u_j. Reading off its coefficients defines the reference-coordinate equivalents:

$$m_j^* = \frac{1}{\phi_{r,j}^2}, \qquad k_j^* = m_j^*\omega_r^2 = \frac{\omega_r^2}{\phi_{r,j}^2}, \qquad c_j^* = 2\zeta_r\omega_r m_j^*, \qquad F_{\mathrm{eq},j}(t) = \frac{Q_r(t)}{\phi_{r,j}}. \tag{3.7}$$

Three consequences deserve emphasis.

First, m_j^* is not a nodal mass. It is the mass an equivalent single oscillator must possess in order that, driven by $F_{\mathrm{eq},j}$, it reproduces the motion of coordinate j. It exceeds or falls short of any physical mass in the structure by whatever factor $\phi_{r,j}^{-2}$ dictates.

Second, the quantities in (3.7) are functions of the pair (r, j). Changing the observation node changes every one of them. The ratio $k_j^*/m_j^* = \omega_r^2$ is the sole coordinate-invariant, which is why frequency agreement is a weaker test than stiffness agreement and why both are reported in Chapter 5.

Third, the scaling of m^* and F_{eq} is consistent by construction: since $m^* \propto \phi^{-2}$, $F_{\mathrm{eq}} \propto \phi^{-1}$

and $u_j \propto \phi$, the product $m^* \ddot{u}_j \propto \phi^{-1}$ matches $F_{\text{eq}} \propto \phi^{-1}$ and the equation balances. Using a resultant force in place of Q_r/ϕ_{ref} breaks exactly this balance, and the resulting error appears wholly in the identified K.

Evaluating (3.7) at the common reference coordinate S_C07 / node 3461 gives the authoritative Phase-2 structural references.

Table 3.3: Reference-coordinate equivalents at S_C07 (node 3461).

Direction	**Mode** r	ϕ_{ref}	f_r **(Hz)**	m^* **(kg)**	k^* **(N m^{-1})**
X	1	$4.0395531850456 \times 10^{-3}$	20.083511908708	61,282.057655069	$9.758260151861 \times 10^{8}$
Y	2	$4.0102126936632 \times 10^{-3}$	20.658102128190	62,182.071430642	$1.047624779593 \times 10^{9}$
Z	4	$4.9242337638946 \times 10^{-3}$	51.434203616507	41,240.381746566	$4.307114855967 \times 10^{9}$

Each row satisfies $m^* = \phi_{\text{ref}}^{-2}$ and $k^* = m^*(2\pi f_r)^2$ to machine precision; these identities were re-derived independently from the stored modal amplitudes and reproduce the tabulated values exactly.

3.4 Generalised Force Projection and the Consistent Right-Hand Side

The load applied in the transient analyses is a distributed resultant force over the top slab face, nominally

$$F_{\text{raw}}(t) \approx 11{,}103 \, \sin(2\pi \cdot 15\, t) \text{ N},$$

acting over a loaded area $A = 20\,\text{m}^2$, equivalent to a peak surface pressure of $p_{\max} = 11{,}103/20 = 555.15\,\text{Pa}$.

The consistent right-hand side of (3.6) is *not* F_{raw}. By (3.4) it is the modal projection $Q_r(t) = \boldsymbol{\phi}_r^{\mathsf{T}} \mathbf{f}(t)$, and by (3.7) it must then be divided by ϕ_{ref}. Because the finite element load vector $\mathbf{f}(t)$ distributes the surface traction over the loaded element faces through the shape functions,

$$\mathbf{f}(t) = p(t) \int_{\Gamma_N} \mathbf{N}^{\mathsf{T}} \mathbf{n} \, \mathrm{d}\Gamma \;\equiv\; p(t)\, \mathbf{f}^{(1\,\text{Pa})},$$

the projection is linear in the pressure history and can be computed once at unit pressure:

$$Q_r(t) = p(t) \underbrace{\boldsymbol{\phi}_r^{\mathsf{T}} \mathbf{f}^{(1\,\text{Pa})}}_{\equiv\, \widehat{Q}_r}, \qquad F_{\text{eq}}(t) = \frac{p(t)\, \widehat{Q}_r}{\phi_{\text{ref}}}. \tag{3.8}$$

The projection was performed over 180 loaded elements on solid face 6, using the solver's own `NOD2SOLV` ordering map so that the modal vector and the load vector are indexed identically — an operation with no approximation in it. The resulting unit-pressure projections and the consequent equivalent-force amplitudes are given in Table 3.4.

The magnitudes of the three ratios are the quantitative justification for Contribution C1. In X and Y the projection is close to unity — the lateral mode shapes are nearly uniform over the loaded face, so the modal participation of a uniform pressure is close to its resultant. In Z

Table 3.4: Exact finite element load-vector projection (180 loaded elements, $A = 20.000\,\mathrm{m}^2$, $p_{\max} = 555.15\,\mathrm{Pa}$).

Direction	**Mode**	$\widehat{Q}_r$ **(N per Pa)**	$F_{\mathrm{eq}}/F_{\mathrm{raw}}$	F_{eq} **peak (N)**
X	1	0.0809163922	1.0015512668524	11,120.223715863
Y	2	0.0808911536	1.0085643802363	11,198.090313764
Z	4	0.0756884952	0.7685306875048	8,532.996223366
Z	7	0.0160809546	—	—

it is 0.7685: the Mode-4 slab shape varies substantially over the loaded face, and 23 % of the applied resultant does not participate in that mode. Had the raw resultant been used as the right-hand side, the identified vertical stiffness would have been inflated by the reciprocal factor, $1/0.7685 = 1.3012$, an error of +30 % — and the resulting fit to the displacement record would have been no worse, because the discrepancy would have been absorbed entirely into K.

The all-mode projection at unit pressure also confirms the modal content of the vertical excitation. Of the ten projected generalised forces, only modes 4 and 7 exceed $10^{-12}\,\mathrm{N\,Pa^{-1}}$; the remaining eight are numerically zero. The vertical load therefore excites exactly two modes, which is precisely the premise of the decomposition in Section 3.5.

3.5 Two-Mode Spatial Decomposition and Modal Filtering

For the lateral directions, (3.5) is an adequate model: the projected force excites one dominant mode and the four fused sensors serve to reduce the influence of residual content. For the vertical direction it is not. The sensor model must be

$$u_j(t) = \phi_{4,j}\, q_4(t) + \phi_{7,j}\, q_7(t), \qquad j = 1, \ldots, s, \tag{3.9}$$

or in matrix form, with $\mathbf{u}(t) \in \mathbb{R}^s$ the vector of simultaneous sensor readings,

$$\mathbf{u}(t) = \boldsymbol{\Phi}\,\mathbf{q}(t), \qquad \boldsymbol{\Phi} = \begin{bmatrix} \phi_{4,1} & \phi_{7,1} \\ \vdots & \vdots \\ \phi_{4,s} & \phi_{7,s} \end{bmatrix} \in \mathbb{R}^{s\times 2}, \qquad \mathbf{q} = \begin{bmatrix} q_4 \\ q_7 \end{bmatrix}. \tag{3.10}$$

With $s = 4 > 2 = p$ the system is overdetermined and the modal coordinates follow from the ordinary least-squares solution

$$\hat{\mathbf{q}}(t) = \left(\boldsymbol{\Phi}^{\mathsf{T}}\boldsymbol{\Phi}\right)^{-1}\boldsymbol{\Phi}^{\mathsf{T}}\mathbf{u}(t) = \boldsymbol{\Phi}^{+}\mathbf{u}(t), \tag{3.11}$$

evaluated independently at each of the 10,000 time samples. The conditioning of this inverse governs how cleanly the two modes separate: for the retained sensor set the modal matrix condition number is

$$\kappa(\boldsymbol{\Phi}) = 1.5340644124117,$$

which is close to unity and indicates that the two mode shapes are well separated over the chosen sensor positions — a direct, quantitative vindication of the placement strategy of Section 1.3. The learning target is then the isolated Mode-4 component referred to the reference coordinate:

$$u_{\mathrm{ref},4}(t) = \phi_{4,\mathrm{S_C07}}\, \hat{q}_4(t). \tag{3.12}$$

The fidelity of (3.11)–(3.12) is established three ways. The four-sensor reconstruction residual is $6.69 \times 10^{-10}\,\%$; the reconstruction at the withheld sensor S_C06 carries an error of $2.99 \times 10^{-9}\,\%$ at a correlation of $1 - 6 \times 10^{-16}$; and the retained modal coordinates satisfy $\mathrm{RMS}(q_7)/\mathrm{RMS}(q_4) = 4.0412\,\%$.

That last figure admits an independent check that shares no intermediate quantity with the decomposition. From (3.4), the steady-state amplitude of mode r under harmonic drive at Ω is $|q_r| = |Q_r|\,|H_r(\Omega)|$ with dynamic compliance

$$H_r(\Omega) = \frac{1}{\omega_r^2 - \Omega^2 + 2\mathrm{i}\zeta_r\omega_r\Omega} \;\approx\; \frac{1}{\omega_r^2 - \Omega^2} \quad \text{for } \zeta_r \ll 1. \tag{3.13}$$

Taking the ratio for modes 7 and 4, and using the projected modal forces

$$\frac{\widehat{Q}_7}{\widehat{Q}_4} = \frac{0.0160809546}{0.0756884952} = 0.212462,$$

with $f_4 = 51.4342\,\mathrm{Hz}$, $f_7 = 114.1081\,\mathrm{Hz}$ and $\Omega = 2\pi \cdot 15\,\mathrm{rad\,s^{-1}}$:

$$\frac{|q_7|}{|q_4|} = \frac{\widehat{Q}_7}{\widehat{Q}_4} \cdot \frac{\omega_4^2 - \Omega^2}{\omega_7^2 - \Omega^2} \implies 4.019\,\%,$$

against the measured $4.041\,\%$. The two routes — one from the assembled finite element load vector and modal frequencies, the other from four independent transient displacement records — agree to a relative $0.55\,\%$. This is the strongest single piece of evidence that the vertical modal decomposition is physically correct rather than merely numerically convenient.

3.6 Spectral Separation and Dynamic Magnification

The excitation is harmonic at $\Omega/2\pi = 15\,\mathrm{Hz}$, below every retained modal frequency. Each reduced oscillator is therefore driven in its stiffness-controlled regime, and the total response is a superposition of a forced component at 15 Hz and a lightly damped transient component at the modal frequency. Figure 3.1 shows this separation directly in the reference-coordinate displacement spectrum.

The presence of both components is what makes the inverse problem well posed. A purely forced response at a single frequency Ω would constrain only the combination $K - m^*\Omega^2$, leaving K and m^* individually indeterminate; the free component at ω_r supplies the second, independent constraint. Since m^* is supplied exogenously from the modal analysis, the dynamic stiffness

$$K_{\mathrm{dyn}}(\Omega) \equiv K - m^*\Omega^2 \tag{3.14}$$

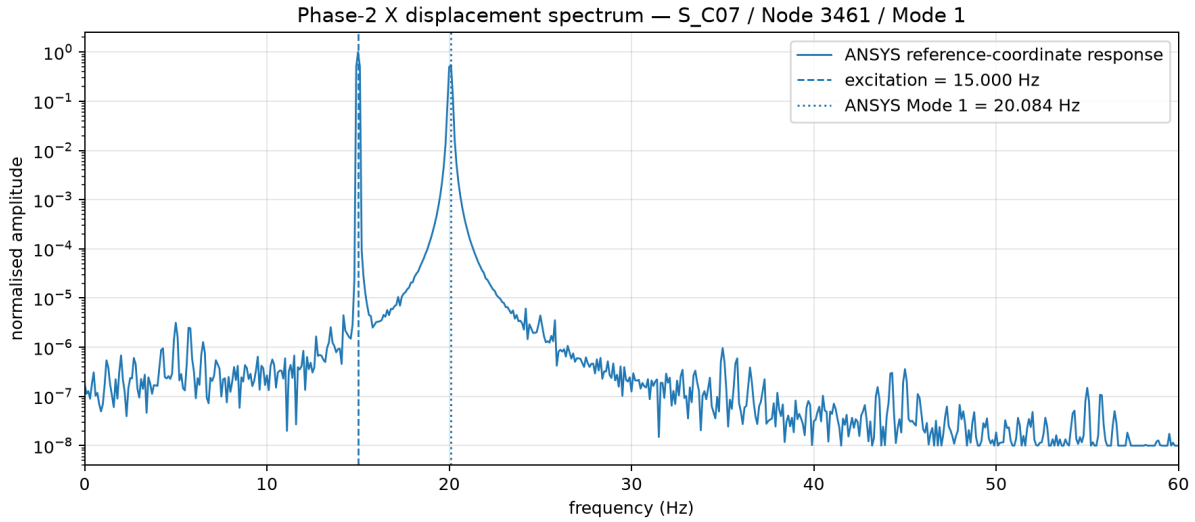


Figure 3.1: Normalised displacement spectrum of the reference-coordinate response, X direction. Two peaks dominate: the forced response at the 15 Hz drive frequency and the free transient at the Mode-1 frequency of 20.084 Hz. The spectral floor lies some six decades below the peaks, confirming that the fused reference coordinate is essentially a two-component signal.

furnishes a scaling-coherence check on the frozen datasets. Evaluated at $\Omega = 2\pi \cdot 15$ before freezing, the relative discrepancies between the reference-coordinate quantities and the recorded response amplitudes were 0.280 % in X, 0.151 % in Y and 0.014 % in Z — an independent confirmation that mass, stiffness, displacement and equivalent force are mutually coherent.

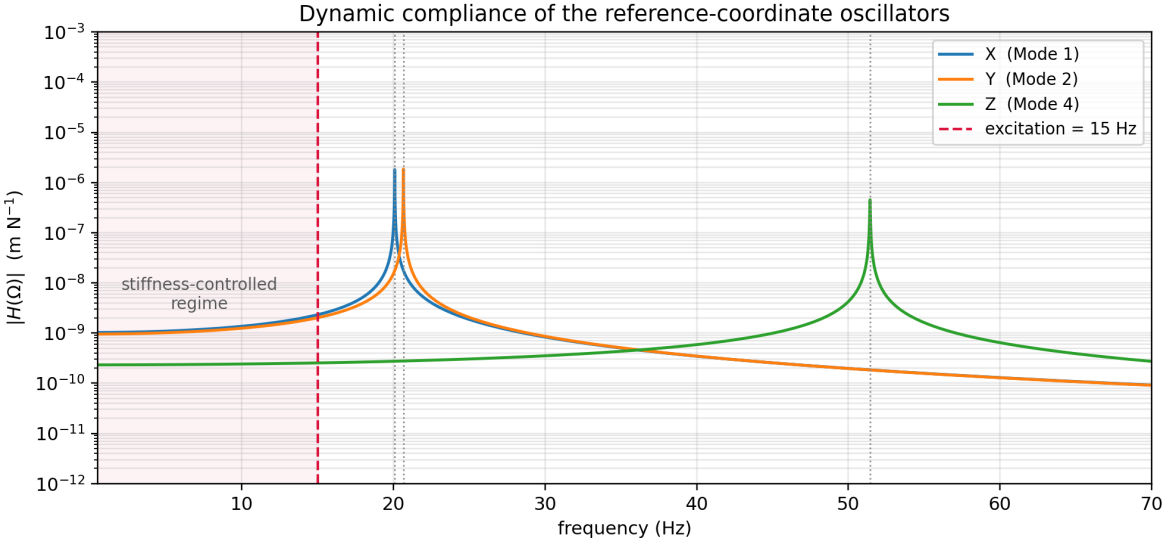


Figure 3.2: Dynamic compliance of the three reference-coordinate oscillators, with the 15 Hz drive line marked. All three systems are driven below resonance, in the stiffness-controlled regime, which is what makes the forced and free spectral components separately visible in Figure 3.1.

3.7 Observability of Stiffness and Damping

The reduced equation contains two unknowns, and they are not equally observable from a finite record. The distinction is physical, not numerical, and is worth deriving.

Write the reduced homogeneous response as $u(t) = A\,\mathrm{e}^{-\zeta\omega_n t}\cos(\omega_d t + \varphi)$ with $\omega_d = \omega_n\sqrt{1-\zeta^2}$. Stiffness enters through $\omega_n = \sqrt{K/m^*}$, which is fixed by the *phase* of the oscillation: over a window of length T containing $N_c = \omega_n T/2\pi$ cycles, a fractional error δ in K produces a cumulative phase error

$$\Delta\varphi = \omega_n T\Big(\sqrt{1+\delta} - 1\Big) \approx \pi N_c\,\delta. \tag{3.15}$$

The sensitivity of the residual to K therefore grows *linearly* with the number of observed cycles. Damping enters through the envelope $\mathrm{e}^{-\zeta\omega_n t}$, whose total excursion over the window is

$$\frac{A(T)}{A(0)} = \mathrm{e}^{-T/\tau}, \qquad \tau = \frac{2m^*}{C}. \tag{3.16}$$

Damping is therefore observable only in proportion to T/τ. The two parameters obey fundamentally different information scalings: stiffness accumulates evidence per *cycle*, damping per *decay time constant*. Table 3.5 evaluates both for the three identified systems using the recovered parameters of Chapter 5.

Table 3.5: Observability diagnostics over the 9.999 s record.

Direction	N_c **cycles observed**	$\tau = 2m^*/C$ **(s)**	T/τ	**Envelope decay** $\mathrm{e}^{-T/\tau}$
X	200.7	33.30	0.300	0.741
Y	206.4	31.93	0.313	0.731
Z	514.1	18.60	0.538	0.584

All three directions observe hundreds of cycles, so stiffness is strongly determined in every case — which is what Chapter 5 confirms. The damping information is likewise present in all three: the free-response envelope falls by 26 %, 27 % and 42 % respectively over the record, well above the threshold at which decay is measurable.

A separate interpretive point must be recorded. The source transient analysis carried **zero physical damping** and a **numerical damping factor of 0.005** in the time-integration scheme. The dissipation visible in the records is therefore algorithmic rather than material. The identified C is accordingly reported throughout as an **effective reduced-model damping coefficient** — the coefficient that reproduces the observed dissipation of this dataset within the reduced model — and is not claimed as a material or structural damping property of steel. This is a statement about the source model, not a limitation of the identification: the network recovers the dissipation that is present, and reports it as such.

3.8 The Neural Surrogate: Approximation and Exact Differentiation

The displacement is represented by a fully connected network $u_\theta : \mathbb{R} \to \mathbb{R}$ with parameters θ, of depth four and width 64, using sinusoidal activations. Its representational adequacy rests on the universal approximation theorems of Cybenko (1989) and Hornik (1991): for any continuous u on a compact interval and any $\epsilon > 0$ there exist finite width and weights with $\sup_t |u_\theta(t) - u(t)| < \epsilon$.

The property actually exploited, however, is not approximation but **differentiability**. Because u_θ is a finite composition of smooth maps, it is C^∞, and its derivatives are available by reverse-mode automatic differentiation as exact derivatives of the surrogate:

$$\dot{u}_\theta(t) = \frac{\partial u_\theta}{\partial t}, \qquad \ddot{u}_\theta(t) = \frac{\partial^2 u_\theta}{\partial t^2}, \tag{3.17}$$

evaluated to floating-point precision at *arbitrary* t, not only at sample times. This is the decisive structural advantage over the classical baseline of Section 2.6. A finite-difference estimate of $\ddot{u}$ from samples spaced Δt apart carries truncation error $O(\Delta t^2 u^{(4)})$ and, more damagingly, amplifies any high-frequency content by Δt^{-2}. The automatic derivative introduces neither error. The identification consequently never requires the velocity and acceleration histories at all — they are retained in the datasets purely as diagnostics.

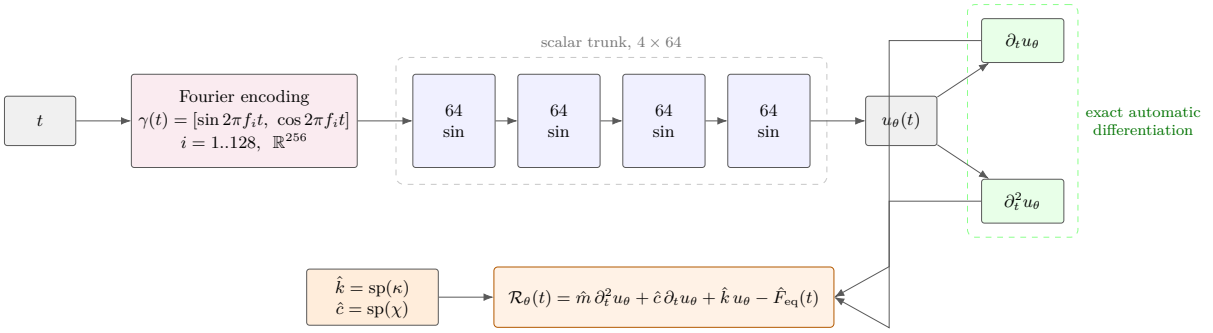


Figure 3.3: Forward architecture of the identification network, showing the fixed Fourier feature encoding, the scalar trunk, and the two automatic-differentiation branches feeding the physics residual. The trainable physical parameters enter only through the residual.

3.9 Spectral Bias and Fourier Feature Encoding

A plain coordinate network fed t directly fails on these records, and the reason is quantitative. By the neural tangent kernel result of Jacot et al. (2018), a wide network trained by gradient descent evolves as

$$\frac{\partial u_\theta(t)}{\partial s} = -\int \mathcal{K}(t, t') \left(u_\theta(t') - u(t')\right) \mathrm{d}t',$$

in training time s, with $\mathcal{K}$ the tangent kernel. Decomposing the error onto the eigenfunctions ψ_i of $\mathcal{K}$ with eigenvalues λ_i, each component decays as $\mathrm{e}^{-\lambda_i s}$. For a plain network the spectrum λ_i falls off rapidly with the frequency of ψ_i, so components at high frequency are suppressed at exponentially slower rates — the spectral bias of Rahaman et al. (2019).

The target here is unambiguously high-frequency: the vertical record carries 514 oscillations of a 51.43 Hz mode within the 9.999 s window. Following Tancik et al. (2020), the remedy is to replace the raw input with a fixed sinusoidal embedding

$$\gamma(t) = \Big[\sin(2\pi f_1 t), \cos(2\pi f_1 t), \ldots, \sin(2\pi f_{n_F} t), \cos(2\pi f_{n_F} t)\Big] \in \mathbb{R}^{2n_F}, \tag{3.18}$$

with f_i spaced linearly over $[f_{\min}, f_{\max}]$. The composition $u_\theta = g_\theta \circ \gamma$ then possesses a tangent kernel that is stationary and whose bandwidth is set by the embedding rather than by depth, so that all frequencies within the encoded band are learned at comparable rates. The present configuration uses $n_F = 128$, giving a 256-dimensional first layer, with $f_{\min} = 5$ Hz throughout, $f_{\max} = 40$ Hz for the lateral directions and $f_{\max} = 60$ Hz for the vertical direction. Both bands bracket the drive frequency and the relevant modal frequency with margin — $[5, 40]$ contains 15 Hz and 20.08/20.66 Hz; $[5, 60]$ contains 15 Hz and 51.43 Hz — which is precisely the design requirement.

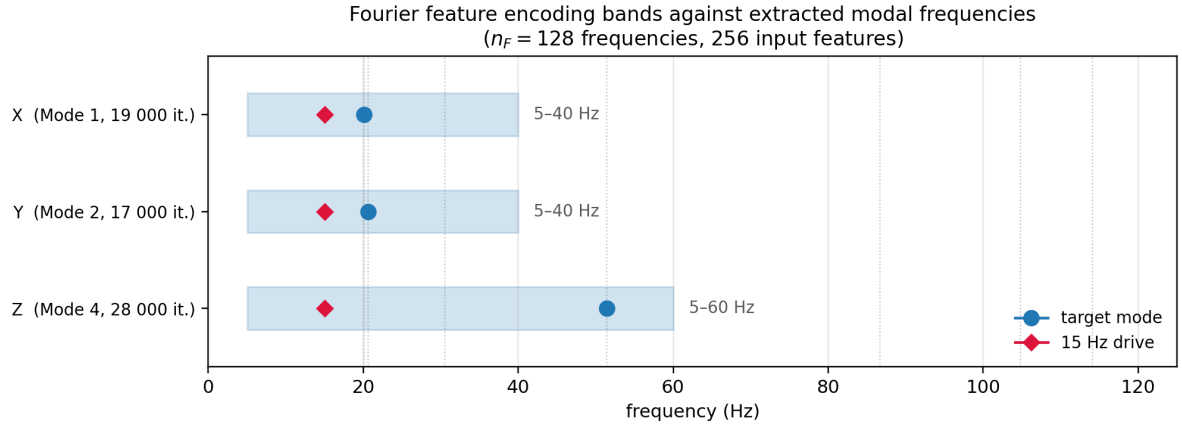


Figure 3.4: Fourier feature encoding bands against the extracted modal frequencies. Each encoded band brackets both the 15 Hz drive frequency and the target modal frequency of its direction, which is the design requirement that the encoding must satisfy.

3.10 The Composite Physics-Informed Loss Functional

Let $\mathcal{D} = \{(t_i, u_i)\}_{i=1}^{N_d}$ denote the retained displacement observations and $\mathcal{T} = \{t_j\}_{j=1}^{N_c}$ the collocation set. Define the physics residual of Section 3.8 evaluated with trainable parameters,

$$\mathcal{R}_\theta(t) = \hat{m}\,\partial_t^2 u_\theta(t) + \hat{c}\,\partial_t u_\theta(t) + \hat{k}\,u_\theta(t) - \hat{F}_{\text{eq}}(t). \tag{3.19}$$

The composite objective is

$$\mathcal{L}(\theta, \kappa, \chi) = \underbrace{\frac{w_{\mathcal{R}}}{N_c}\sum_{j=1}^{N_c} \big|\mathcal{R}_\theta(t_j)\big|^2}_{\text{physics}} + \underbrace{\frac{w_{\mathcal{D}}}{N_d}\sum_{i=1}^{N_d} \big|u_\theta(t_i) - u_i\big|^2}_{\text{data}}, \tag{3.20}$$

with $w_{\mathcal{R}} = 1$ and $w_{\mathcal{D}} = 100$, minimised jointly over the network parameters θ *and* the physical parameters (κ, χ) underlying $(\hat{k}, \hat{c})$.

The mechanism by which (3.20) identifies parameters deserves to be stated precisely, because it is frequently mischaracterised. The data term alone is satisfied by any surrogate that interpolates the observations, and is entirely indifferent to $\hat{k}$ and $\hat{c}$ — their gradients through that term are identically zero. The identification is effected wholly by the physics term. Writing the stationarity condition for $\hat{k}$:

$$\frac{\partial \mathcal{L}}{\partial \hat{k}} = \frac{2 w_{\mathcal{R}}}{N_c} \sum_j \mathcal{R}_\theta(t_j)\, u_\theta(t_j) = 0 \quad \Longrightarrow \quad \langle \mathcal{R}_\theta,\, u_\theta \rangle_{\mathcal{T}} = 0, \tag{3.21}$$

and correspondingly $\langle \mathcal{R}_\theta, \partial_t u_\theta \rangle_{\mathcal{T}} = 0$ for $\hat{c}$. At convergence the residual is therefore orthogonal, over the collocation set, to both the displacement surrogate and its first derivative. These are exactly the normal equations of a least-squares fit of the equation of motion — but formed against the *network's own smooth derivatives* rather than against differenced data. The weighting $w_{\mathcal{D}}/w_{\mathcal{R}} = 100$ is the statement that the displacement record is trusted more strongly than the residual at any individual collocation point, which pins the surrogate to the data while leaving the residual free to determine the parameters.

Two structural consequences follow. First, the parameters are identified from the *whole* collocation set, including the 20 % of time samples withheld from $\mathcal{D}$; the physics term is evaluated everywhere. Second, a network that merely memorised the displacement record would leave $\hat{k}$ and $\hat{c}$ near their initial values, because memorisation drives the data term to zero without engaging (3.21) at all. Convergence of the parameters away from initialisation is therefore itself evidence that the physics term is doing the work — a point developed empirically in Chapter 5.

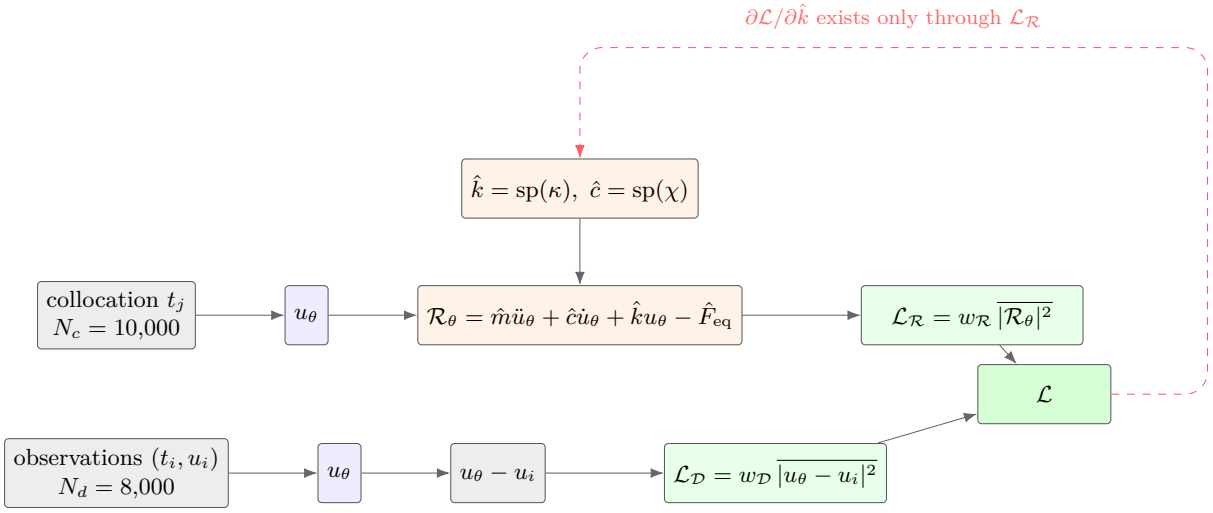


Figure 3.5: Computational graph of the composite objective. The trainable physical parameters are reachable by gradient only through the physics branch; the data branch cannot influence them.

3.11 Non-Dimensionalisation and Parameter Scaling

The physical quantities span many orders of magnitude — displacements of order $10^{-5}\,\mathrm{m}$ against stiffnesses of order $10^{9}\,\mathrm{N\,m^{-1}}$ — which is numerically intolerable for gradient-based optimisation in single precision. Introduce reference scales U_0 for displacement and F_0 for force, and set $u = U_0\hat{u}$, $F = F_0\hat{F}$. Substituting into the reduced equation and dividing throughout by F_0:

$$\frac{m^* U_0}{F_0}\,\ddot{\hat{u}} + \frac{C\,U_0}{F_0}\,\dot{\hat{u}} + \frac{K\,U_0}{F_0}\,\hat{u} = \hat{F}(t),$$

which defines the scaled parameter group

$$\hat{m} = \frac{m^* U_0}{F_0}, \qquad \hat{c} = \frac{C\,U_0}{F_0}, \qquad \hat{k} = \frac{K\,U_0}{F_0}, \tag{3.22}$$

with the inverse map $K = \hat{k}F_0/U_0$, $C = \hat{c}F_0/U_0$. **Time is deliberately left unscaled**, in physical seconds, so that every automatic derivative in (3.17) is a derivative with respect to real time and the recovered $\hat{c}$ carries the correct dimensions without a further correction. Taking F_0 as the equivalent-force peak and U_0 as the order of the reference displacement gives $\hat{k}$ of order unity — for the X case, $\hat{k}_{\mathrm{ref}} = 8.775$; for Z, 1.010 — which is the intended conditioning.

Table 3.6: Scaling constants and resulting reference magnitudes.

Direction	U_0 **(m)**	F_0 **(N)**	$\hat{m}$	$\hat{k}_{\mathrm{ref}} = k^* U_0/F_0$
X	1.0×10^{-4}	11,120.223715863	5.510866×10^{-4}	8.775237262485
Y	1.0×10^{-4}	11,198.090313764	5.552917×10^{-4}	9.355387840595
Z	2.0×10^{-6}	8,532.996223366	9.666096×10^{-6}	1.009519925527

3.12 Positivity Enforcement by Softplus Reparameterisation

Stiffness and damping of a stable passive structure are strictly positive. Left unconstrained, a gradient step can carry either into the negative half-line, at which point the reduced system is non-physical — negative C implies energy injection, negative K implies static instability — and the optimisation may stall in a region with no physical meaning. This is not hypothetical: the classical least-squares diagnostic of Section 2.6, run on the same data without constraint, returns negative damping in all three directions.

Rather than impose an inequality constraint, the parameters are reparameterised through the softplus map

$$\hat{k} = \mathrm{sp}(\kappa) = \log\!\left(1 + \mathrm{e}^{\kappa}\right), \qquad \hat{c} = \mathrm{sp}(\chi), \qquad \kappa, \chi \in \mathbb{R}, \tag{3.23}$$

so that the optimiser works on unconstrained variables while the physical parameters are positive by construction. The map is smooth, strictly increasing, and has derivative

$$\frac{\mathrm{d}\,\mathrm{sp}}{\mathrm{d}\kappa} = \sigma(\kappa) = \frac{1}{1 + \mathrm{e}^{-\kappa}} \in (0, 1),$$

so gradients propagate without saturation for moderate κ and are gently damped as $\kappa \to -\infty$, which prevents the parameter from being driven towards zero by a single large step. Initialisation uses the exact inverse $\kappa_0 = \mathrm{sp}^{-1}(\hat{k}_0) = \log(\mathrm{e}^{\hat{k}_0} - 1)$, so that the stated initial values $\hat{k}_0 = 1.0$ and $\hat{c}_0 \in \{10^{-4}, 2 \times 10^{-7}\}$ are realised precisely rather than approximately.

3.13 Joint Optimisation by Adam

The objective (3.20) is minimised over the concatenated parameter vector $\Theta = (\theta, \kappa, \chi)$ by the adaptive-moment method of Kingma and Ba (2015). With $g_s = \nabla_\Theta \mathcal{L}$ at step s:

$$\begin{aligned} m_s &= \beta_1 m_{s-1} + (1-\beta_1) g_s, & \hat{m}_s &= m_s/(1-\beta_1^s), \\ v_s &= \beta_2 v_{s-1} + (1-\beta_2) g_s^{\odot 2}, & \hat{v}_s &= v_s/(1-\beta_2^s), \\ \Theta_{s+1} &= \Theta_s - \eta\, \hat{m}_s \oslash (\sqrt{\hat{v}_s} + \varepsilon), \end{aligned} \tag{3.24}$$

with $\eta = 10^{-3}$ and the standard $\beta_1 = 0.9$, $\beta_2 = 0.999$. The per-coordinate normalisation by $\sqrt{\hat{v}_s}$ is what makes joint optimisation over θ and (κ, χ) tractable: the gradients with respect to a single scalar physical parameter, accumulated over 10,000 collocation points, differ in magnitude from those with respect to an individual weight by several orders, and an unnormalised method would advance one while leaving the other essentially static. Adam rescales both to comparable effective step sizes, which is why the parameters traverse the full distance from initialisation to their converged values — a factor of 8.76 in $\hat{k}$ for the X case — within the allotted iteration budget.

The trajectories themselves are recorded every 100 iterations and are presented, with the corresponding validated stopping points, in Chapter 5.

Chapter 4

Methodology

4.1 Computational Environment and Execution

All training was executed on a Slurm-managed GPU partition of the university high-performance computing cluster, accessed over SSH from a local development environment. The runtime stack is PyTorch with DeepXDE configured on the PyTorch backend; the physics residual, the force interpolator and the trainable parameters are all expressed as PyTorch tensors so that reverse-mode differentiation propagates through the entire computational graph, including the interpolated forcing.

Each direction was submitted as an independent batch job. The three production runs are summarised in Table 4.1. Total wall-clock cost for the complete three-direction identification was 844.48 s — a little over fourteen minutes — at an essentially constant cost per iteration of 13.1–13.4 ms, confirming that the per-step cost is set by the network and collocation size rather than by any direction-specific property.

Table 4.1: Production run configuration and computational cost.

Quantity	**X**	**Y**	**Z**
Validated iterations	19,000	17,000	28,000
Wall-clock runtime (s)	253.822	223.270	367.385
Cost per iteration (ms)	13.359	13.133	13.121
Random seed	0	0	0
Optimiser	Adam	Adam	Adam
Learning rate	1×10^{-3}	1×10^{-3}	1×10^{-3}
Collocation points N_c	10,000	10,000	10,000
Observations N_d	8,000	8,000	8,000
Withheld time samples	2,000	2,000	2,000

4.2 Software Architecture and the Direction-Agnostic Configuration Path

The implementation is a single Python package, `sysid`, comprising six modules with strictly separated responsibilities: configuration resolution and validation; data ingestion and contract checking; physics, scaling and the residual construction; training; evaluation; and visualisation. There is no direction-specific code path anywhere in the package. The direction is selected by a single `active_direction` key in one YAML file, and the loader then merges a common block with a direction-specific `cases` block, applying `network_overrides` and `training_overrides` where present. Output routing to `outputs/phase2/{X,Y,Z}` is likewise derived from the active direction, which prevents concurrent cluster jobs from overwriting one another.

This is not a convenience. It is the mechanism by which the claim of uniformity in Chapter 5 is made verifiable: the X, Y and Z results were produced by the same network topology, the same residual, the same loss weighting, the same optimiser and the same evaluation code, differing only in the four quantities that *must* differ — the input dataset, the reference modal quantities, the non-dimensionalisation scales, and the encoding bandwidth.

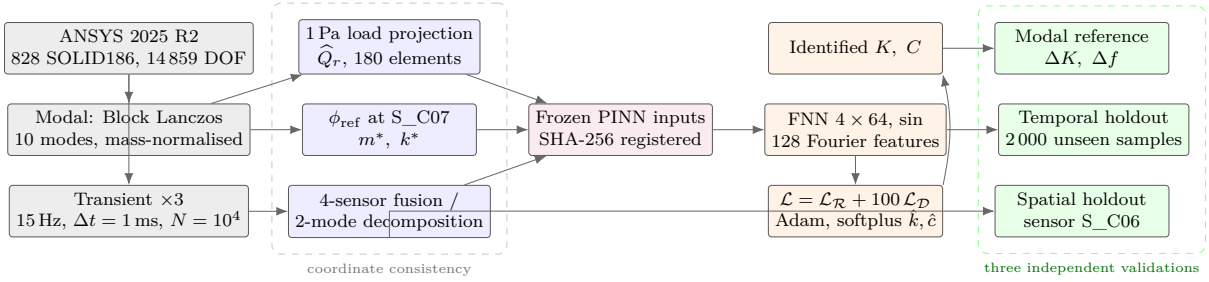


Figure 4.1: End-to-end Phase-2 pipeline, from the finite element solves through the coordinate-consistency preprocessing to the identified parameters and their three independent validations.

4.3 Finite Element Source: Modal and Transient Analyses

The modal analysis (system A5) was solved by block Lanczos iteration for ten modes with mass normalisation, no prestress and no physical damping, yielding the frequencies of Table 3.2 and the mass-normalised mode shapes from which every reference-coordinate quantity in this chapter derives.

Three independent transient analyses (system B5) were then solved, one per excitation direction, with force configurations $[F_X(t), 0, 0]$, $[0, F_Y(t), 0]$ and $[0, 0, F_Z(t)]$. In each, a distributed resultant face force of nominal peak 11,103 N at 15 Hz was applied over the 20 m^2 top slab face. Solution points were stored from $t = 0.001$ s to $t = 10.000$ s at $\Delta t = 0.001$ s, giving exactly $N = 10{,}000$ samples per record. No synthetic $t = 0$ sample was inserted at any stage; the datasets begin where the solver's stored output begins.

Two properties of the source solution are carried forward as interpretive constraints. First, the physical damping matrix is zero. Second, the time-integration scheme employs a numerical damping factor of 0.005. The dissipation present in the records is therefore algorithmic, and

the identified C is reported throughout as an *effective reduced-model damping coefficient* rather than as a material property, as established in Section 3.7.

The original force table was specified on approximately 0.01 s knots while the transient solution was stored at 0.001 s, so the extracted force history carries the solver's own tabular interpolation behaviour. This history was used exactly as exported; it was not replaced by an analytic sine, because doing so would introduce a forcing the solver never applied and would transfer the discrepancy into the identified parameters.

4.4 Observation Node Selection and Roles

Eight nodes were selected on the top slab surface (Figure 1.1, Table 4.2) under five rules: slab nodes only, no leg nodes; avoid edges and corners; avoid slab–leg interface regions; distribute spatially over the slab; and retain at least one node for spatial holdout. The same eight nodes serve all three directions, so that the spatial basis is identical across the experiment.

Table 4.2: Observation nodes and final roles.

Sensor	**Node**	x **(m)**	y **(m)**	z **(m)**	**Role**
S_C01	5003	2.2914	6.3760	0.35	excluded from fusion
S_C02	3514	4.4581	6.0427	0.35	training fusion
S_C03	3507	6.7914	6.0427	0.35	excluded from fusion
S_C04	3408	2.4581	3.3760	0.35	excluded from fusion
S_C05	3402	4.4581	3.3760	0.35	training fusion
S_C06	3396	6.4581	3.3760	0.35	**hidden spatial validation**
S_C07	**3461**	3.4581	4.7093	0.35	**fusion — reference coordinate**
S_C08	3454	5.7914	4.7093	0.35	training fusion

The three exclusions are not arbitrary. A directional screening of the eight candidates against the target modes returned a normalised vertical score of 0.175, 0.175 and 0.039 for S_C04, S_C03 and S_C01 respectively — an order of magnitude below the retained sensors — and flagged all three with weak Z observability. Retaining them would have degraded the conditioning of the vertical modal matrix in (3.10) without contributing independent information, and they were removed on that basis before any training took place.

4.5 Multi-Sensor Reference-Coordinate Fusion (X and Y)

For the lateral directions the four retained sensors carry a single dominant modal contribution, and the modal amplitudes at those sensors are nearly equal in magnitude (Table 4.3). The

four simultaneous records are mapped to the common modal coordinate and reconstructed at the reference node under (3.5), producing one scalar reference-coordinate observation per time sample.

Table 4.3: Mass-normalised modal amplitudes at the fusion sensors.

Sensor	ϕ_1 **(X, Mode 1)**	ϕ_2 **(Y, Mode 2)**
S_C02	$3.974164254243 \times 10^{-3}$	$4.059136687694 \times 10^{-3}$
S_C05	$3.974164254358 \times 10^{-3}$	$4.059136687691 \times 10^{-3}$
S_C07 (ref.)	$4.039553185046 \times 10^{-3}$	$4.010212693663 \times 10^{-3}$
S_C08	$4.039553185118 \times 10^{-3}$	$4.010212693700 \times 10^{-3}$

The fusion residual — the discrepancy between the four measured records and their single-mode reconstruction — is 0.01924 % in X and 0.02863 % in Y. Reconstructed against the directly measured reference-node record, the fused signal agrees to 0.01893 % (X) and 0.02898 % (Y) at correlations of 0.99999999 and 0.99999997. The fusion therefore does not distort the reference response; it suppresses residual content while preserving the dominant modal signal.

4.6 Two-Mode Spatial Decomposition (Z)

The vertical case is handled differently, for the reason established in Section 3.5: the slab motion carries both a Mode-4 and a Mode-7 contribution, and the all-mode load projection confirms that the vertical excitation drives exactly those two modes and no others. The local modal diagnostic at the reference sensor makes the necessity explicit — at S_C07 the Mode-4 amplitude accounts for a squared fraction of only 0.3540 of the local response, and the locally largest mode is in fact Mode 6. A naive single-mode treatment of the raw vertical record at this node would therefore have identified the wrong mode.

The four-sensor overdetermined system (3.10)–(3.11) is solved instead at each of the 10,000 samples, with modal matrix condition number $\kappa(\mathbf{\Phi}) = 1.534$. The retained learning target is the isolated Mode-4 component referred to S_C07, equation (3.12). The resulting record differs from the directly measured vertical displacement at that node by 2.747 %, which is the quantitative measure of the Mode-7 content that has been removed.

Three fidelity measures confirm the separation: a four-sensor reconstruction residual of 6.691×10^{-10} %, a hidden-sensor reconstruction error of 2.986×10^{-9} %, and a modal amplitude ratio $\mathrm{RMS}(q_7)/\mathrm{RMS}(q_4) = 4.041$ % that is independently predicted to 0.55 % by the route of Section 3.5.

4.7 Hidden-Sensor Spatial Validation

Sensor S_C06 was withheld from every fusion and decomposition operation. Its finite element record was extracted identically to the training sensors and then set aside. The modal coordinate estimated from S_C02, S_C05, S_C07 and S_C08 alone is used to predict the response at S_C06, and that prediction is compared against the withheld record.

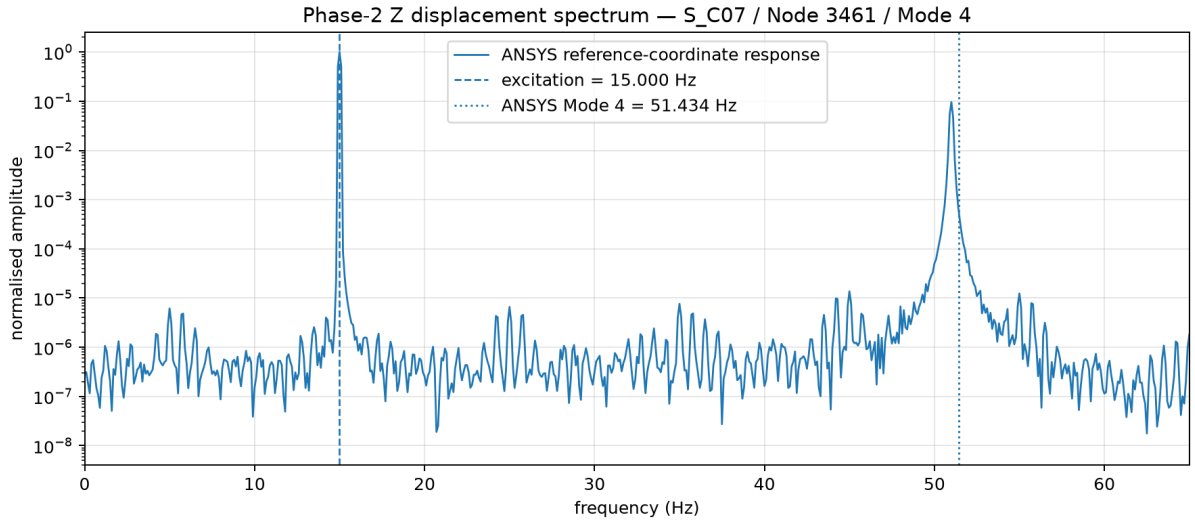


Figure 4.2: Normalised spectrum of the isolated Mode-4 reference-coordinate record, Z direction. The two-mode decomposition has left a clean two-component signal: the 15 Hz forced response and the 51.434 Hz Mode-4 transient.

This is a test of the *spatial* modal hypothesis: it asks whether the modal coordinate recovered from four locations correctly predicts the motion at a fifth location it never saw. It is conceptually and numerically distinct from the temporal train/holdout split described in Section 4.11, and the two are never combined into a single figure of merit. Results are given in Section 5.2.

4.8 Exact Load-Vector Projection and the Equivalent Force

The equivalent force was constructed by the procedure of Section 3.4 and not by any scalar heuristic. A unit surface pressure was assembled over the loaded face, producing a finite element load vector over 180 loaded elements on solid face 6 with a total loaded area of 20.000 m^2. That vector was projected onto each mass-normalised mode using the solver's `NOD2SOLV` node-ordering map, guaranteeing that the modal vector and the load vector are indexed consistently. The resulting unit-pressure generalised forces $\widehat{Q}_r$ (Table 3.4) were then scaled by the physical pressure history, $p_{\max} = 555.15\,\mathrm{Pa}$, and divided by ϕ_{ref} to yield the reference-coordinate equivalent force.

The all-mode projection at unit pressure is itself a verification: for the vertical load, modes 4 and 7 return 7.569×10^{-2} and $1.608 \times 10^{-2}\,\mathrm{N\,Pa^{-1}}$ while the remaining eight modes return magnitudes between 1.6×10^{-14} and $2.1 \times 10^{-12}\,\mathrm{N\,Pa^{-1}}$ — numerically zero. The two-mode premise is therefore not an assumption but a measured property of the load vector.

The original Workbench model files were not modified at any point in this operation. The raw face-force columns are retained in the final datasets for provenance only and are never used as the physics right-hand side.

4.9 Dataset Freezing, Quality Control and Provenance

Three final datasets were assembled, one per direction, each containing time, reference-coordinate displacement, reference-coordinate velocity and acceleration, the raw face force and the equivalent force. Each carries exactly 10,000 samples on the identical time grid $t \in [0.001, 10.000]\,\mathrm{s}$, contains no non-finite values, and was verified against the data contract in the configuration: expected sample count, expected time step to $10^{-10}\,\mathrm{s}$, expected endpoints, and scalar agreement to a relative tolerance of 10^{-9}. These tolerances are deliberately tight because the datasets are frozen simulation output rather than field measurements; any deviation indicates a pipeline fault rather than measurement scatter.

Each frozen dataset was registered by SHA-256 digest, and the training pipeline never writes to them. A final frequency-domain coherence check was applied before freezing, comparing the recorded response amplitudes against the dynamic stiffness (3.14) at the 15 Hz drive; relative discrepancies were 0.280 % (X), 0.151 % (Y) and 0.014 % (Z), confirming that mass, stiffness, displacement and equivalent force are mutually coherent.

4.10 Network Construction and Fourier Encoding as Implemented

The network is a fully connected map with layer widths $[1, 64, 64, 64, 64, 1]$, sinusoidal activations and Glorot uniform initialisation. Before the first layer, the scalar input is replaced by the Fourier embedding (3.18) with $n_F = 128$ frequencies spaced linearly over the direction-specific band, which expands the effective input dimension to 256. The embedding frequencies are fixed constants, not trainable, and are moved to the active device and dtype at evaluation time so that the encoding participates correctly in the autograd graph.

The equivalent force enters the residual through a differentiable piecewise-linear interpolator built once from the frozen CSV history. DeepXDE evaluates the residual at arbitrary collocation times rather than only at the stored sample times, so interpolation is unavoidable; the implementation caches the constant time and force tables per dtype/device pair and performs a linear interpolation that is exact at the stored samples. No sine-wave approximation is introduced at any point, and the interpolator rejects any query outside the stored interval rather than extrapolating.

4.11 Collocation, Holdout and the Displacement-Only Data Term

The time domain $[0.001, 10.000]\,\mathrm{s}$ carries 10,000 collocation points at which the physics residual is enforced. The displacement observations are drawn from the same frozen record, but every fifth sample is withheld from the data term, giving $N_d = 8{,}000$ training observations and 2,000 held-out samples distributed uniformly through the record rather than clustered at one end. The held-out samples are never presented to the data term at any iteration.

Table 4.4: Network and encoding configuration.

Quantity	X	Y	Z
Layer widths	$[1, 64, 64, 64, 64, 1]$	$[1, 64, 64, 64, 64, 1]$	$[1, 64, 64, 64, 64, 1]$
Activation	sin	sin	sin
Initialiser	Glorot uniform	Glorot uniform	Glorot uniform
Fourier frequencies n_F	128	128	128
Encoding band (Hz)	5–40	5–40	5–60
Effective input dimension	256	256	256
Target mode within band	20.084 Hz	20.658 Hz	51.434 Hz
Drive frequency within band	15 Hz	15 Hz	15 Hz

Two properties of this construction should be noted. First, the physics residual is still evaluated over the full collocation set, including the time locations of held-out samples; the withholding applies to the *observation* term only, which is what makes the subsequent holdout error a genuine test of interpolation rather than a test of whether the residual was enforced. Second, and decisively, **velocity and acceleration are never supplied to the network**. The observation term contains displacement alone. The velocity and acceleration columns present in the datasets are used exclusively for preprocessing quality control, derivative consistency checks and the classical diagnostic of Section 4.13.

4.12 Non-Dimensionalisation, Positivity and Optimiser Configuration

The scaling of Section 3.11 is applied with the constants of Table 3.6, leaving time in physical seconds. Both trainable parameters are held under the softplus reparameterisation (3.23), initialised through the exact inverse map so that the nominal initial values are realised precisely. Initialisation is $\hat{k}_0 = 1.0$ in all three directions; $\hat{c}_0 = 1 \times 10^{-4}$ for the lateral directions and 2×10^{-7} for the vertical direction, the latter reflecting the different force and displacement scales in that case.

The composite loss (3.20) is compiled with weights $[w_{\mathcal{R}}, w_{\mathcal{D}}] = [1, 100]$ and both scalar parameters registered as externally trainable variables so that Adam updates them alongside the network weights. Parameter values are recorded every 100 iterations to a convergence trace, giving 191 recorded states for the X run, 171 for Y and 281 for Z.

4.13 The Classical Least-Squares Diagnostic

A linear least-squares estimator is run alongside each identification, not as a competing method but as a coherence check on the frozen data. Assembling the design matrix from the exported

velocity and acceleration columns,

$$\underbrace{\begin{bmatrix} u(t_1) & \dot{u}(t_1) \\ \vdots & \vdots \\ u(t_N) & \dot{u}(t_N) \end{bmatrix}}_{\mathbf{A}} \begin{bmatrix} K \\ C \end{bmatrix} = \underbrace{\begin{bmatrix} F_{\mathrm{eq}}(t_1) - m^*\ddot{u}(t_1) \\ \vdots \\ F_{\mathrm{eq}}(t_N) - m^*\ddot{u}(t_N) \end{bmatrix}}_{\mathbf{b}}, \tag{4.1}$$

and solving $\min_{K,C} \|\mathbf{A}[K, C]^{\mathsf{T}} - \mathbf{b}\|_2$ yields an independent stiffness estimate from a route that shares no machinery with the network. The estimator is unconstrained, so it is free to return negative damping — and, as Section 5.7 shows, it does so in every direction. That behaviour is the diagnostic's most informative output, and the reason it is designated a diagnostic rather than a reference.

4.14 Reproducibility Controls

The following controls were applied. The random seed is fixed and recorded in every output artefact. The frozen datasets are SHA-256 registered and are opened read-only by the pipeline. Every run writes an identification record containing the resolved configuration, the fixed inputs, the reference modal quantities, the scaled and physical parameters, the run metadata, the fit statistics and the validation statistics, so that each reported number is traceable to the run that produced it. The production iteration counts are stored in the configuration rather than passed at the command line, so a production run cannot silently use a different budget. Output routing is derived from the active direction, so concurrent jobs cannot overwrite one another's results.

Chapter 5

Results and Discussion

5.1 Overview

The identification succeeded in all three directions. Table 5.1 is the consolidated result; the remainder of the chapter establishes each entry and examines why the outcome is as uniform as it is.

Table 5.1: Consolidated Phase-2 identification result, reference coordinate S_C07 / node 3461.

Quantity	**X (Mode 1)**	**Y (Mode 2)**	**Z (Mode 4)**
Equivalent mass m^* (kg)	61,282.058	62,182.071	41,240.382
Equivalent force peak (N)	11,120.224	11,198.090	8,532.996
Identified K ($\mathbf{N\,m^{-1}}$)	$\mathbf{9.741138 \times 10^8}$	$\mathbf{1.045455 \times 10^9}$	$\mathbf{4.303617 \times 10^9}$
ANSYS modal k^* ($\mathrm{N\,m^{-1}}$)	9.758260×10^8	1.047625×10^9	4.307115×10^9
ΔK vs modal reference	**−0.175463 %**	**−0.207090 %**	**−0.081206 %**
Identified C ($\mathrm{N\,s\,m^{-1}}$)	3,681.130	3,894.639	4,433.640
Derived f_n (Hz)	20.065885	20.636701	51.413315
Modal f_r (Hz)	20.083512	20.658102	51.434204
Δf_n vs modal reference	**−0.087770 %**	**−0.103599 %**	**−0.040611 %**
Derived ζ (%)	0.023822	0.024152	0.016640
Holdout RMSE (m)	5.268964×10^{-8}	6.160486×10^{-8}	3.563753×10^{-9}
Holdout relative RMSE	**0.263542 %**	**0.355181 %**	**0.248078 %**
Holdout R^2	**0.999993055**	**0.999987385**	**0.999993846**

Three observations frame the discussion. The stiffness errors span 0.081 %–0.207 %, a range of a factor of 2.6 across three structurally dissimilar modes whose frequencies differ by a factor of 2.6 and whose equivalent masses differ by 51 %. The holdout coefficients of determination agree to the fifth decimal place across all three. The vertical case returned the smallest stiffness deviation

of the three, but that figure must not be read as evidence that the two-mode decomposition produced superior accuracy. As Section 5.8 establishes, the vertical non-dimensionalisation happens to place the reference at $\hat{k}_{\text{ref}} = 1.0095$ against the global initialisation $\hat{k}_0 = 1.0$, so that run began 0.94 % from its answer and traversed almost no parameter space. The vertical number is a scaling artefact as much as a result, and is treated as such throughout this chapter.

5.2 Preprocessing Fidelity: Spatial Validation at the Withheld Sensor

Before any statement about the identified parameters can be made, the reference-coordinate observation must be shown to be a faithful representation of the structure's motion. Table 5.2 reports the spatial validation.

Table 5.2: Preprocessing fidelity and spatial validation at the withheld sensor S_C06.

Quantity	**X**	**Y**	**Z**
Participating modes retained	1	1	2
Training reconstruction NRMSE (%)	1.9240×10^{-2}	2.8626×10^{-2}	6.6907×10^{-10}
Hidden S_C06 error (%)	$\mathbf{9.7392 \times 10^{-3}}$	$\mathbf{1.5588 \times 10^{-1}}$	$\mathbf{2.9859 \times 10^{-9}}$
Hidden S_C06 R^2	0.9999999905	0.9999975700	—
Hidden S_C06 correlation	0.9999999985	0.9999988634	$1 - 6 \times 10^{-16}$
Fused vs direct reference NRMSE (%)	1.8926×10^{-2}	2.8979×10^{-2}	—
Velocity derivative correlation	0.99999958	0.99999905	0.99994173
Acceleration derivative correlation	0.99838755	0.99816176	0.98748659
Modal matrix condition $\kappa(\mathbf{\Phi})$	—	—	1.5341

The modal coordinate estimated from four sensors predicts the motion at a fifth, entirely unseen sensor to within 0.0097 % in X, 0.1559 % in Y and 3.0×10^{-9} % in Z. The vertical figure is not a typographical error: because the vertical response is *exactly* a two-mode superposition — as the all-mode load projection independently establishes — the four-sensor system in (3.10) is not merely overdetermined but exactly consistent, and the least-squares solution recovers the modal coordinates to machine precision. The lateral cases carry residual content from non-retained modes and therefore reconstruct to finite, though still very small, error.

The Y case is the weakest of the three, by an order of magnitude in both the fusion residual and the hidden-sensor error. Section 5.10 establishes why, and shows that the same cause propagates consistently through every subsequent Y diagnostic.

5.3 Independent Verification of the Vertical Modal Decomposition

The vertical decomposition rests on the claim that exactly two modes participate and that their amplitudes are correctly separated. Two entirely independent routes were used to test that claim, sharing no intermediate quantity.

The *first* route is the transient displacement decomposition itself, which returns

$$\left[\frac{\mathrm{RMS}(q_7)}{\mathrm{RMS}(q_4)}\right]_{\text{measured}} = 4.0412\,\%.$$

The *second* route uses only the finite element load vector and the modal frequencies. From the unit-pressure projections,

$$\frac{\widehat{Q}_7}{\widehat{Q}_4} = \frac{1.60809546 \times 10^{-2}}{7.56884952 \times 10^{-2}} = 0.212462;$$

combining with the dynamic compliance ratio (3.13) at $\Omega = 2\pi \cdot 15\,\mathrm{rad\,s^{-1}}$ with $f_4 = 51.4342\,\mathrm{Hz}$ and $f_7 = 114.1081\,\mathrm{Hz}$ gives

$$\left[\frac{|q_7|}{|q_4|}\right]_{\text{predicted}} = 4.019\,\%.$$

The relative discrepancy is 0.55 %. One route begins from assembled load vectors and eigenvalues; the other from four transient displacement records. Their agreement to better than one part in 180 establishes that the vertical modal content has been correctly identified and correctly separated, and it does so without reference to the network at any point.

A secondary measure supports the same conclusion: the Mode-7 contribution to the raw reference-node response has a root-mean-square ratio of 2.823 % relative to Mode 4, and removing it changes the reference record by 2.747 %. These two figures are consistent with one another and with the 4.041 % modal-coordinate ratio once the differing modal amplitudes at S_C07 are accounted for.

5.4 Surrogate Accuracy and Temporal Generalisation

Table 5.3 reports the accuracy of the displacement surrogate, separating the fit over all samples from the fit over the 2,000 time samples withheld from the data term.

The decisive line is the holdout-to-training ratio. A surrogate that had memorised the observations would fit them far better than the withheld samples, and the ratio would be substantially greater than unity. The measured ratios are 1.00070, 1.00183 and 1.02364 — that is, the error at time locations the data term never saw exceeds the error at the observed locations by 0.07 %, 0.18 % and 2.4 % respectively. The surrogate is interpolating a continuous function, not fitting a point set.

The maximum absolute errors are likewise informative: at $1.86 \times 10^{-7}\,\mathrm{m}$, $2.04 \times 10^{-7}\,\mathrm{m}$ and

Table 5.3: Surrogate accuracy and temporal holdout validation.

Quantity	X	Y	Z
RMS of reference record (m)	1.999241×10^{-5}	1.734389×10^{-5}	1.435876×10^{-6}
Peak reference displacement (m)	4.118887×10^{-5}	3.623133×10^{-5}	2.643622×10^{-6}
RMSE, all samples (m)	5.266004×10^{-8}	6.151502×10^{-8}	3.498061×10^{-9}
Relative RMSE, all samples (%)	0.263400	0.354678	0.243619
Maximum absolute error (m)	1.858616×10^{-7}	2.039350×10^{-7}	1.291333×10^{-8}
R^2, all samples	0.999993062	0.999987420	0.999994065
RMSE, training samples (m)	5.265263×10^{-8}	6.149254×10^{-8}	3.481444×10^{-9}
RMSE, held-out samples (m)	$\mathbf{5.268964 \times 10^{-8}}$	$\mathbf{6.160486 \times 10^{-8}}$	$\mathbf{3.563753 \times 10^{-9}}$
Holdout / training RMSE ratio	**1.00070**	**1.00183**	**1.02364**
Relative holdout RMSE (%)	0.263542	0.355181	0.248078
Holdout R^2	0.999993055	0.999987385	0.999993846

1.29×10^{-8} m they are 0.45 %, 0.56 % and 0.49 % of the corresponding peak displacements, and are distributed through the record rather than concentrated at the endpoints. Figures 5.1–5.3 show this directly.

5.5 Identified Parameters

The identified physical parameters, together with the scaled quantities actually optimised and the derived structural descriptors, are collected in Table 5.4.

Table 5.4: Identified parameters, scaled and physical.

Quantity	X	Y	Z
$\hat{k}$ (optimised)	8.759840012	9.336013794	1.008700132
$\hat{k}_{\mathrm{ref}}$ (modal)	8.775237262	9.355387841	1.009519926
$\hat{c}$ (optimised)	3.310302×10^{-5}	3.477950×10^{-5}	1.039175×10^{-6}
$\hat{m}$ (fixed)	5.510866×10^{-4}	5.552917×10^{-4}	9.666096×10^{-6}
K ($\mathrm{N\,m^{-1}}$)	9.741138×10^{8}	1.045455×10^{9}	4.303617×10^{9}
C ($\mathrm{N\,s\,m^{-1}}$)	3.681130×10^{3}	3.894639×10^{3}	4.433640×10^{3}
ω_n ($\mathrm{rad\,s^{-1}}$)	126.077672	129.664214	323.039388
f_n (Hz)	20.065885	20.636701	51.413315
ζ	2.382208×10^{-4}	2.415194×10^{-4}	1.663996×10^{-4}
$\tau = 2m^*/C$ (s)	33.295	31.932	18.603
Oscillations observed	200.7	206.4	514.1

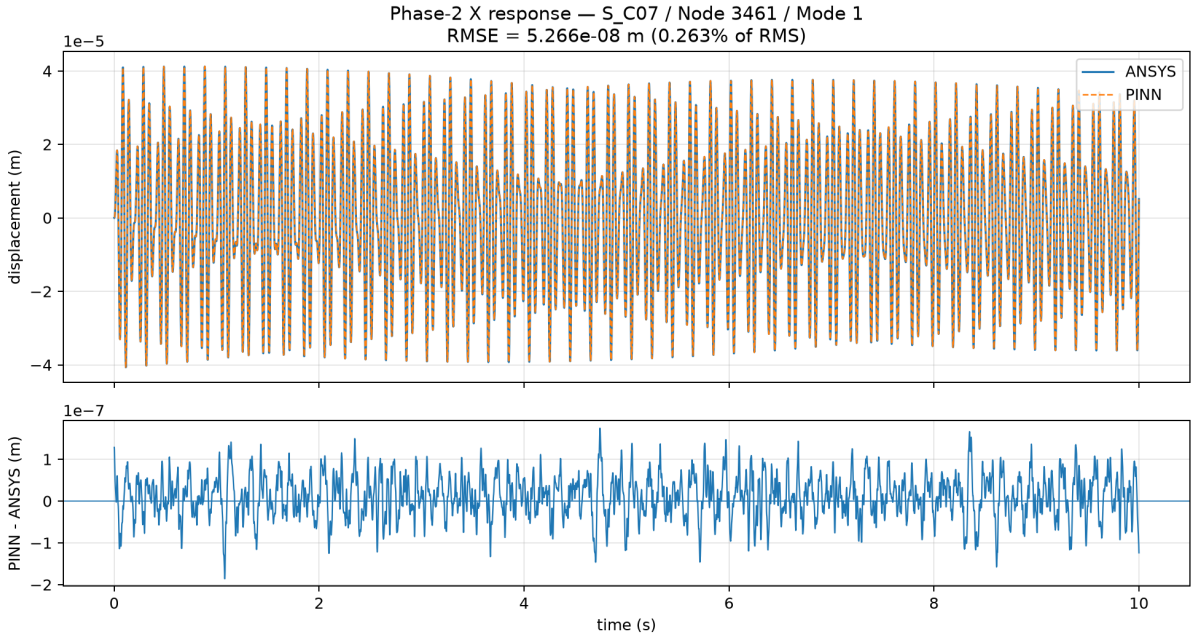


Figure 5.1: Predicted against reference-coordinate displacement, X direction, over the full 9.999 s record (approximately 201 oscillations of the Mode-1 response). The upper panel overlays the two traces; the lower panel shows the signed difference, bounded by 1.86×10^{-7} m against a peak displacement of 4.12×10^{-5} m.

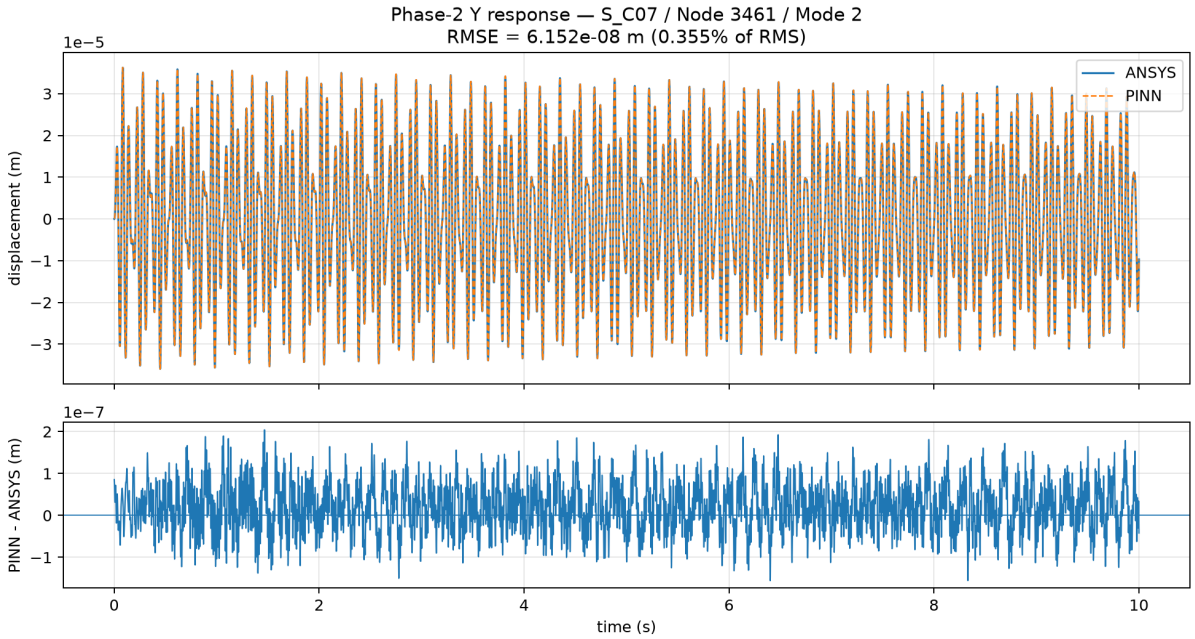


Figure 5.2: Predicted against reference-coordinate displacement, Y direction.

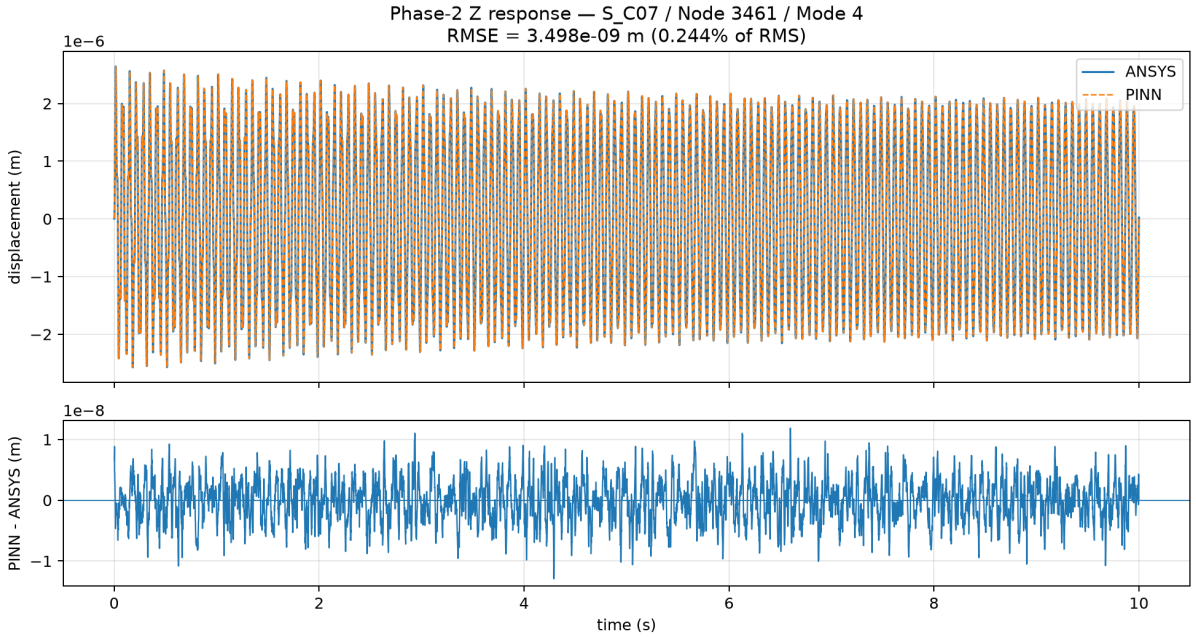


Figure 5.3: Predicted against the isolated Mode-4 reference-coordinate displacement, Z direction, over approximately 514 oscillations. The 128-frequency encoding over 5–60 Hz sustains the fit across the entire record without spectral degradation at late times.

The scaled stiffnesses are all of order unity — 8.76, 9.34 and 1.01 — confirming that the non-dimensionalisation achieved its conditioning objective. The derived damping ratios fall in the narrow band 0.017 %–0.024 %, which is both physically plausible for the algorithmic dissipation of a lightly damped time-integration scheme and consistent across directions.

5.6 Agreement with the Authoritative Modal Reference

The authoritative Phase-2 reference for each direction is the reference-coordinate equivalent stiffness computed from the mass-normalised modal analysis at S_C07, equation (3.7). This is the correct benchmark because it is derived from the same coordinate to which the observation, the mass and the force have all been referred. Table 5.5 gives the comparison; Figures 5.4–5.6 show it.

Two structural features of this table deserve comment.

First, the frequency error is in every case the stiffness error halved, together with its exact second-order correction. Since $f_n \propto K^{1/2}$ at fixed m^*, a relative perturbation δ in K produces

$$\frac{\Delta f_n}{f_n} = \sqrt{1+\delta} - 1 = \frac{\delta}{2} - \frac{\delta^2}{8} + O(\delta^3).$$

Evaluating for X with $\delta = -1.754625 \times 10^{-3}$ gives $\delta/2 = -8.77313 \times 10^{-4}$ and $-\delta^2/8 = -3.848 \times 10^{-7}$, summing to -8.77698×10^{-4}, against a measured -8.77698×10^{-4}. The corresponding sums for Y and Z reproduce -1.035985×10^{-3} and -4.061136×10^{-4} to seven significant figures. This exact reproduction confirms that m^* was held fixed at its modal value throughout and that no mass drift entered the optimisation — the entire frequency deviation is

Table 5.5: Agreement with the ANSYS modal reference at S_C07.

Quantity	**X**	**Y**	**Z**
PINN K ($\mathrm{N\,m^{-1}}$)	9.741138×10^8	1.045455×10^9	4.303617×10^9
Modal k^* ($\mathrm{N\,m^{-1}}$)	9.758260×10^8	1.047625×10^9	4.307115×10^9
Stiffness ratio K/k^*	0.99824537	0.99792910	0.99918794
Stiffness difference (%)	**−0.175463**	**−0.207090**	**−0.081206**
PINN f_n (Hz)	20.065885	20.636701	51.413315
Modal f_r (Hz)	20.083512	20.658102	51.434204
Frequency ratio	0.99912230	0.99896401	0.99959389
Frequency difference (%)	**−0.087770**	**−0.103599**	**−0.040611**

attributable to the stiffness deviation and to nothing else.

Second, all three errors are negative. The identified stiffness is systematically 0.08 %–0.21 % *below* the modal reference in every direction. A systematic bias of consistent sign across three independent runs is not random scatter; it reflects the small amount of energy the reduced model must account for that the single retained mode does not carry, which the residual absorbs as a marginally reduced effective stiffness. The magnitude of that bias tracks the modal purity of each case, as Section 5.10 demonstrates.

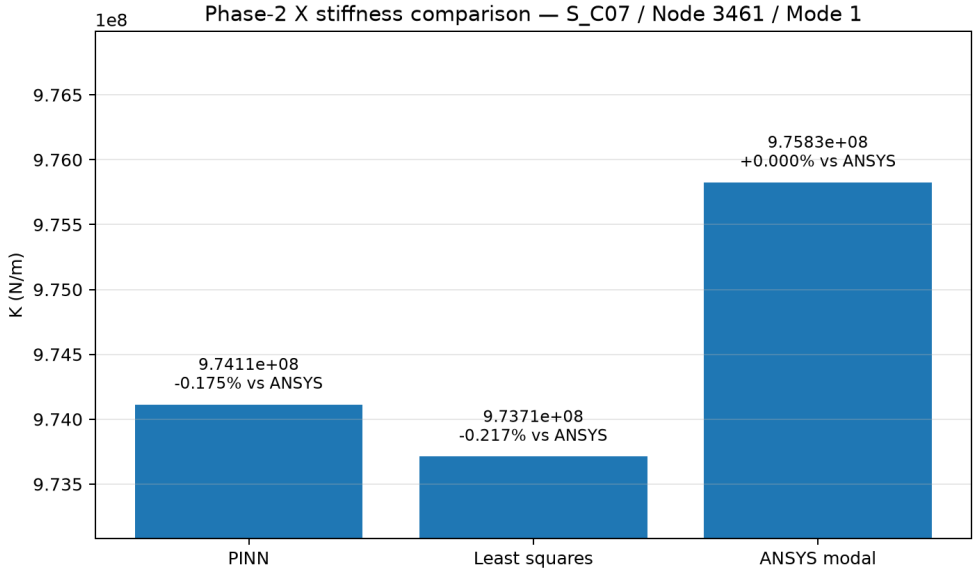


Figure 5.4: Identified stiffness by method, X direction. The PINN and the least-squares diagnostic bracket one another within 0.042 %, and both sit 0.18–0.22 % below the ANSYS modal reference. Note the suppressed vertical axis: the full range of the plot spans 0.35 % of the stiffness value.

5.7 Agreement with the Classical Least-Squares Diagnostic

The unconstrained least-squares estimator of Section 4.13 provides a stiffness estimate from a route that shares no machinery with the network — different derivatives, different objective,

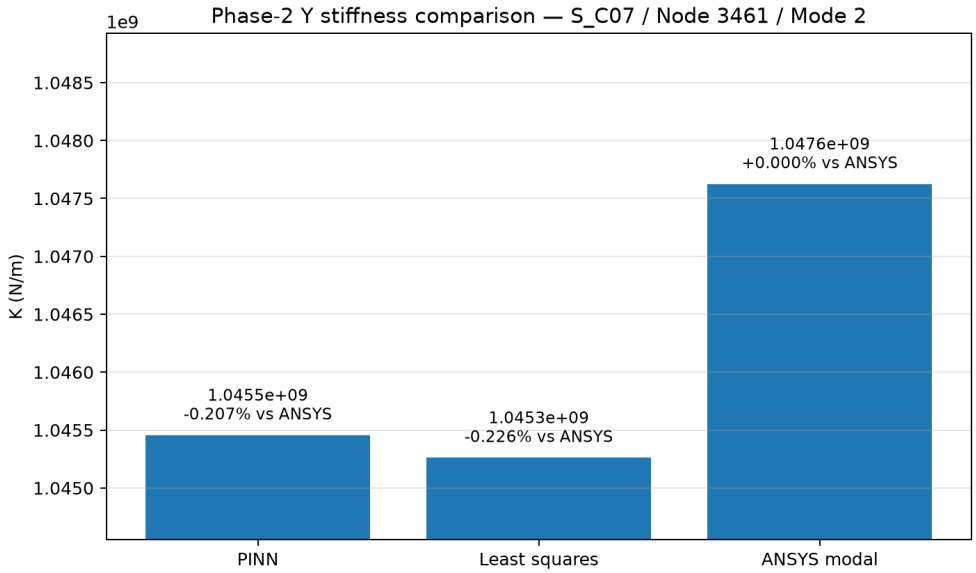


Figure 5.5: Identified stiffness by method, Y direction.

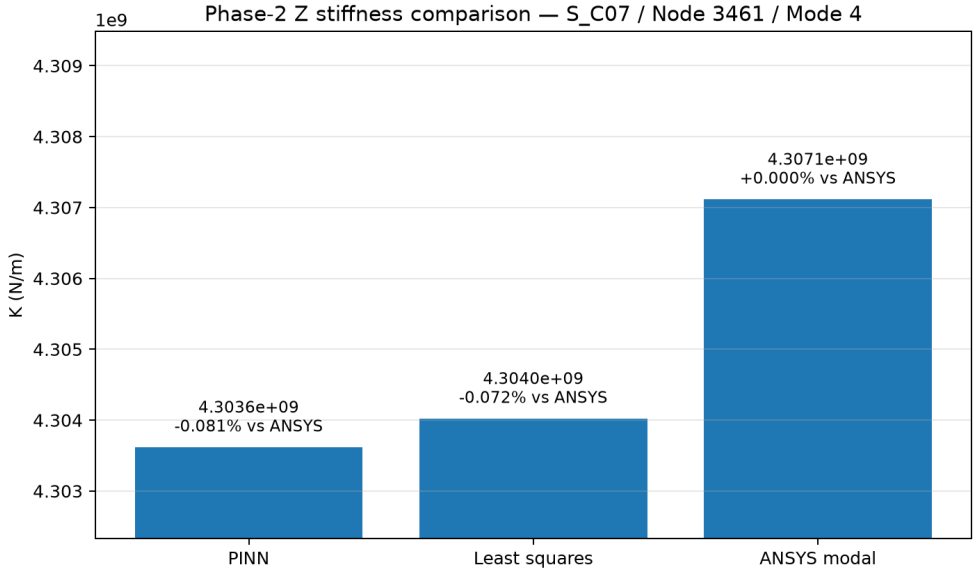


Figure 5.6: Identified stiffness by method, Z direction.

different solver. Table 5.6 reports it.

Two results must be read together here, and the second is uncomfortable.

The stiffness agreement is exact for practical purposes: 0.041 %, 0.019 % and 0.009 %. Two estimators that share nothing but the input data converge on the same stiffness to better than one part in two thousand in the worst case and one part in ten thousand in the best. That is strong evidence that the identified K is a property of the data rather than an artefact of either estimator.

It is also, unavoidably, evidence that the network earns nothing on stiffness. The least-squares estimator is not a weak baseline that the network improves upon; it is an exact closed-form solution of the same two-parameter problem, and Table 5.7 gives its cost.

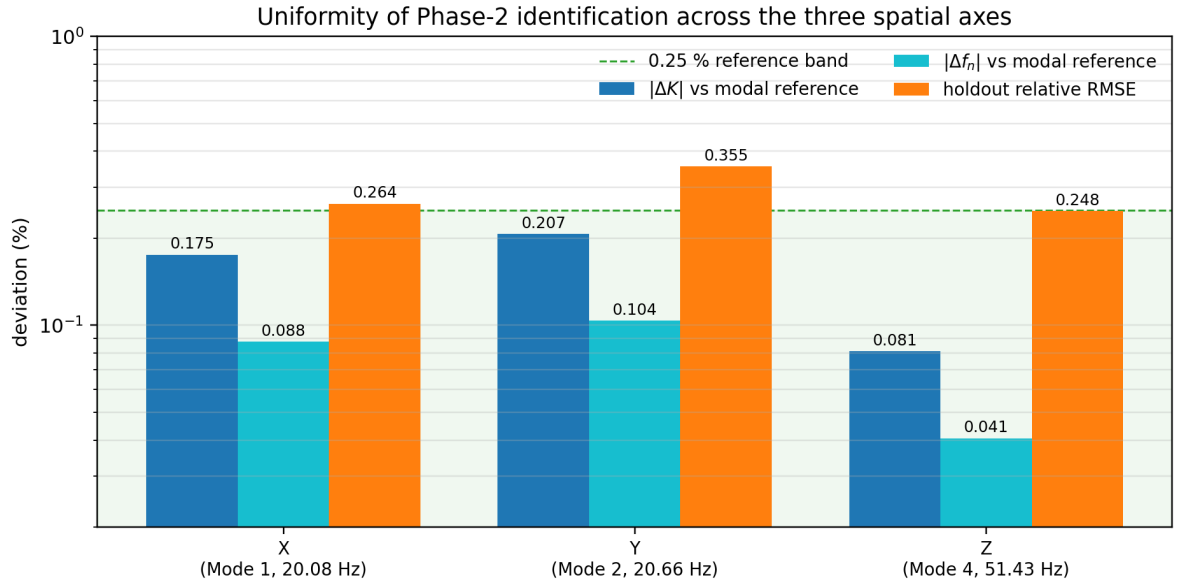


Figure 5.7: Uniformity of the Phase-2 identification across the three spatial axes: absolute stiffness and frequency deviation from the modal reference, and holdout relative RMSE, on a logarithmic scale against a 0.25 % reference band.

Table 5.6: Classical least-squares diagnostic.

Quantity	**X**	**Y**	**Z**
LS K ($\mathrm{N\,m^{-1}}$)	9.737121×10^8	1.045261×10^9	4.304021×10^9
PINN vs LS, ΔK (%)	**+0.041258**	**+0.018628**	**−0.009383**
LS f_n (Hz)	20.061747	20.634779	51.415728
LS C ($\mathrm{N\,s\,m^{-1}}$)	-3.561515×10^5	-3.728526×10^5	-5.670111×10^5
LS ζ (%)	−2.305275	−2.312396	−2.127958
Force reconstruction R^2	0.99893959	0.99363096	0.99936397
Relative force RMSE (%)	3.256388	7.980626	2.521970
Design matrix rank	2	2	2
Design matrix condition $\kappa(\mathbf{A})$	104.244	104.831	103.641

The network is approximately 10^6 times more expensive and returns the same stiffness. No argument from accuracy, convergence quality or generalisation can recover that gap, and none is attempted here. If stiffness were the only quantity of interest in this reduced setting, ordinary least squares would be the correct method and the network would be an elaborate way of reproducing it.

The damping result is where the two estimators part company, and it is the whole of the network's case. The least-squares estimator returns *negative* damping in all three directions, at ratios near −2.1 % to −2.3 % of critical. A negative damping coefficient implies energy injection over the record and is physically impossible for a passive steel structure. The estimate is not merely imprecise; it is inadmissible, and no post-processing recovers a usable number from it.

Table 5.7: Computational cost of the two estimators on the identical frozen datasets. The least-squares timing is the mean of 200 repeated solves of the rank-2 normal equations over all 10,000 samples on a single CPU core; the network timing is the recorded wall-clock of the validated production run on a cluster GPU. The comparison is of end-to-end cost to obtain (K, C), not of like-for-like hardware.

Quantity	**X**	**Y**	**Z**
Least squares, time to solution (ms)	0.224	0.183	0.179
PINN, time to solution (s)	253.822	223.270	367.385
Cost ratio, PINN : LS	$\mathbf{1.1 \times 10^6}$	$\mathbf{1.2 \times 10^6}$	$\mathbf{2.1 \times 10^6}$
$\lvert\Delta K\rvert$, PINN vs LS (%)	0.041	0.019	0.009
LS damping ζ (%)	−2.305	−2.312	−2.128
PINN damping ζ (%)	+0.0238	+0.0242	+0.0166

The cause is not ill-conditioning. The design matrix has full rank 2 and a condition number near 10^2 in every direction, which is thoroughly well conditioned; the normal equations are solved to machine precision. The cause is the input. The estimator must be supplied with $\dot{u}$ and $\ddot{u}$, and the only available sources are the solver's exported derivative columns — discrete time-integration quantities that already carry the scheme's own numerical dissipation — or finite differences of the displacement record, which amplify high-frequency content by Δt^{-2}. Either route injects the dissipation signature into the very regressor whose coefficient is meant to measure dissipation. Regressed against a model that attributes all damping to $C\dot{u}$, the estimator absorbs the discrepancy by reversing the sign.

The network cannot fail this way, because it never performs the operation. Only displacement enters the data term; $\dot{u}_\theta$ and $\ddot{u}_\theta$ are exact derivatives of a continuous surrogate, obtained by automatic differentiation at collocation points, and the softplus reparameterisation forbids a negative value from being reached at all. The result is a positive, physically admissible damping coefficient in all three directions, of consistent magnitude.

This is the narrow ground on which the method is defended, and it should be stated without inflation. The physics-informed formulation is not recommended here as a general replacement for ordinary least squares in a reduced single-degree-of-freedom setting: on stiffness it is 10^6 times more expensive for no gain. It is recommended as a *derivative-free estimator*, admissible in exactly the regime where the derivative-consuming alternative breaks — when the dissipation to be identified is of the same order as the artefacts introduced by estimating the derivatives. On this dataset that regime is not hypothetical; it is where the classical estimator actually sits, and the boundary is Table 5.7's last two rows.

5.8 Parameter Convergence and Validated Stopping

Table 5.8 summarises the parameter trajectories recorded every 100 iterations; Figures 5.8–5.10 show them.

Table 5.8: Parameter convergence and stability.

Quantity	X	Y	Z
K at initialisation ($\mathrm{N\,m^{-1}}$)	1.112022×10^8	1.119809×10^8	4.266498×10^9
K at stop ($\mathrm{N\,m^{-1}}$)	9.741138×10^8	1.045455×10^9	4.303617×10^9
K **traverse factor**	**8.760×**	**9.336×**	1.009×
C at initialisation ($\mathrm{N\,s\,m^{-1}}$)	1.112023×10^4	1.119809×10^4	8.532993×10^2
C at stop ($\mathrm{N\,s\,m^{-1}}$)	3.681131×10^3	3.894639×10^3	4.433640×10^3
C **traverse factor**	**0.331×**	**0.348×**	**5.196×**
First iteration within 1 % of final K	11,200	12,000	0
K range over final 10 % of iterations (%)	0.0407	0.0170	0.0087
C range over final 10 % of iterations (%)	1.166	0.817	16.953
K range over final 5 % of iterations (%)	0.0407	0.0117	0.0072
C range over final 5 % of iterations (%)	0.661	0.677	9.042
Recorded convergence states	191	171	281

The stiffness trajectories are monotone and settle onto stable plateaus. In X and Y the parameter traverses factors of 8.76 and 9.34 from its initialisation, reaching within 1 % of its final value at iterations 11,200 and 12,000 respectively, and thereafter varying by less than 0.05 % over the remaining iterations. The validated stopping points of 19,000 and 17,000 therefore allow 7,800 and 5,000 iterations of confirmed plateau beyond first convergence.

Retraction: the vertical stiffness result is a scaling artefact

The vertical case requires an explicit retraction of a claim that an earlier reading of these results would support, and that this chapter does not.

The vertical non-dimensionalisation scale is $U_0 = 2 \times 10^{-6}\,\mathrm{m}$. Combined with the equivalent-force peak $F_0 = 8{,}532.996\,\mathrm{N}$, equation (3.22) places the modal reference at

$$\hat{k}_{\mathrm{ref}} = \frac{k^* U_0}{F_0} = \frac{4.307115 \times 10^9 \times 2 \times 10^{-6}}{8{,}532.996} = 1.009520,$$

against a global initialisation of $\hat{k}_0 = 1.0$ applied identically in all three directions. The vertical run therefore began $(1.009520 - 1.0)/1.009520 = 0.94\,\%$ from its own target, and finished 0.081 % from it. Its entire stiffness trajectory spans a traverse factor of 1.009×, against 8.760× in X and

9.336× in Y, and Table 5.8 records that it was already within 1 % of its final value at iteration zero.

Three consequences follow, and all three are concessions.

The vertical result is not evidence that the two-mode decomposition produced superior accuracy. Any estimator initialised 0.94 % from the answer, on a well-conditioned problem, would report a small final deviation. The decomposition is justified by the diagnostics in Sections 5.2 and the independent Q_7/Q_4 cross-check — a reconstruction error of 3.0×10^{-9} % at the withheld sensor and a 0.55 % agreement between two unrelated routes — and not by the stiffness figure, which cannot carry that weight and is withdrawn from that role here.

The three directions are therefore not directly comparable as convergence demonstrations. X and Y traversed nearly an order of magnitude in $\hat{k}$ and constitute genuine convergence from an uninformed start; Z did not. Ranking the three by final stiffness deviation compares two convergence results against one near-stationary one.

The choice of U_0 was made for numerical conditioning, not to flatter the result, and that is exactly why it went unexamined: a scale chosen so that $\hat{k}$ is of order unity will, whenever the reference itself is near unity in those units, place the initialisation near the answer by coincidence. The lesson is a methodological one — a non-dimensionalisation chosen for conditioning must be checked against the initialisation it implies — and it applies to any future direction added to this framework.

What survives is narrower. The vertical run does move, and it moves toward the reference rather than away from it, settling to a final-window stability of 0.0087 %. Its damping coefficient traverses a factor of 5.196 from initialisation, so the run is not stationary in parameter space; it is the stiffness scale, not the optimisation, that begins close to the answer. The evidence against memorisation in Section 5.11 is accordingly carried by the X and Y stiffness traverses and by the damping traverse in all three directions, and explicitly not by the vertical stiffness.

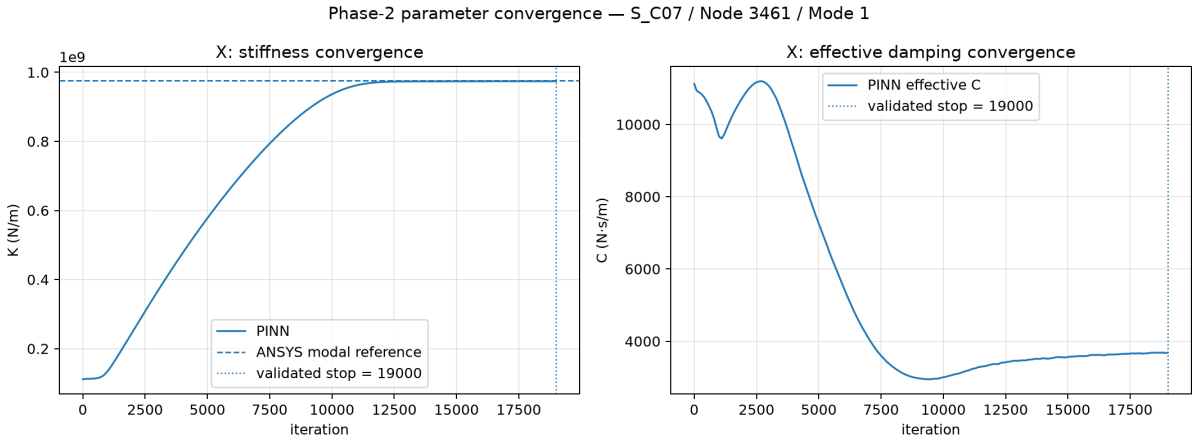


Figure 5.8: Stiffness and effective damping convergence, X direction, with the ANSYS modal reference line and the validated stopping point at 19,000 iterations. The stiffness rises monotonically onto the reference and remains there; the damping coefficient passes through a transient excursion before settling.

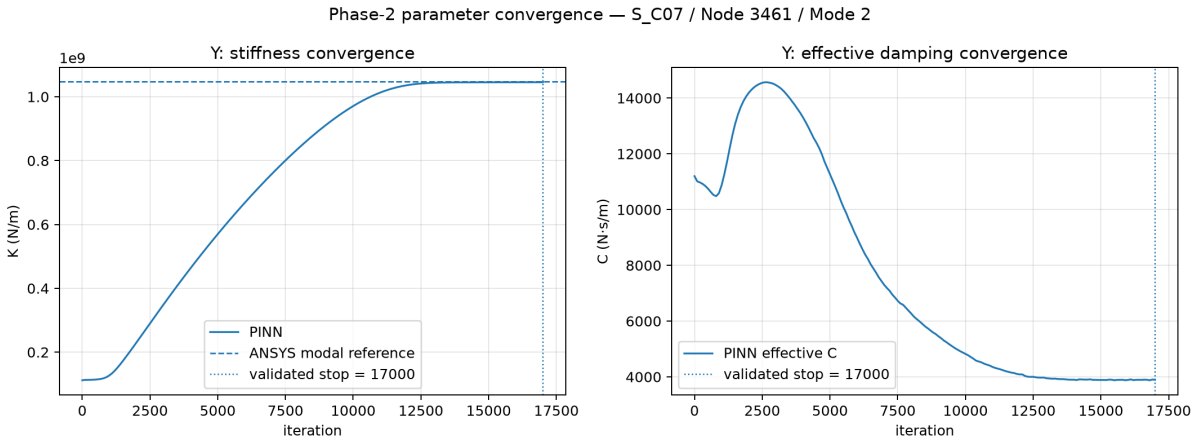


Figure 5.9: Stiffness and effective damping convergence, Y direction, validated stop at 17,000 iterations.

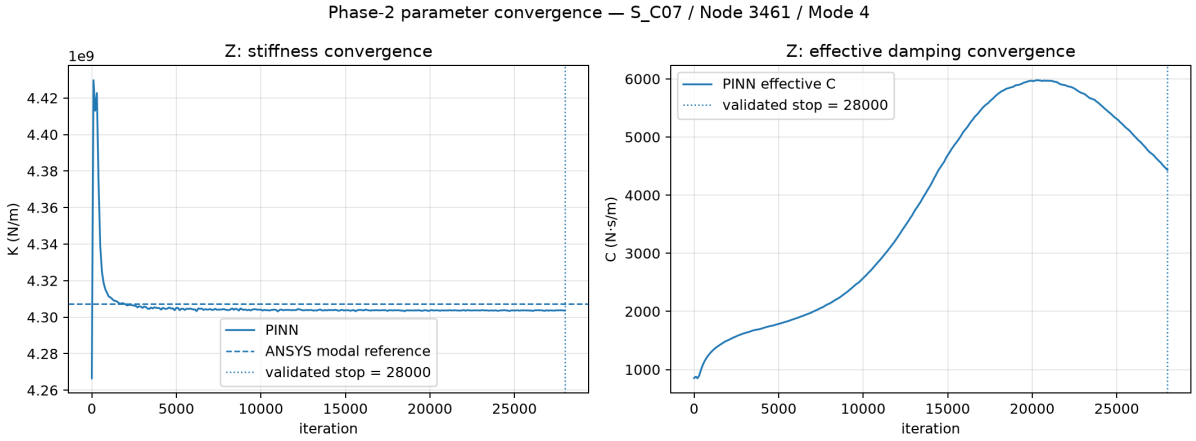


Figure 5.10: Stiffness and effective damping convergence, Z direction, validated stop at 28,000 iterations. The stiffness plateau is the tightest of the three at 0.0087 % over the final 2,800 iterations; the damping coefficient climbs by a factor of 5.2 from its initialisation.

5.9 Energy Audit and the Cost of Numerical Differentiation

An energy audit of the identified models provides an additional, physically motivated consistency check, and simultaneously quantifies the penalty that numerical differentiation imposes. Over the record, conservation requires

$$\underbrace{\int_0^T F_{\mathrm{eq}}\dot{u}\,\mathrm{d}t}_{W_{\mathrm{in}}} = \underbrace{\int_0^T C\dot{u}^2\,\mathrm{d}t}_{E_{\mathrm{diss}}} + \underbrace{\left[\tfrac{1}{2}m^*\dot{u}^2 + \tfrac{1}{2}Ku^2\right]_0^T}_{\Delta E_{\mathrm{stored}}}. \tag{5.1}$$

Table 5.9 evaluates (5.1) using the identified parameters together with velocities and accelerations obtained by **central differencing of the measured record** — deliberately, so that the audit exercises the classical derivative route rather than the network's.

The energy balance closes to 1.6 % in X and 5.0 % in Y, which is satisfactory given that both sides of (5.1) are formed from differenced data. The vertical closure error of 15.3 % looks anomalous until it is read alongside the last two rows. The vertical acceleration derivative correlation is

Table 5.9: Energy audit of the identified models (finite-difference derivatives).

Quantity	**X**	**Y**	**Z**
Work input W_{in} (J)	5.779472×10^{-1}	3.545917×10^{-1}	1.992936×10^{-3}
Dissipated E_{diss} (J)	1.589260×10^{-1}	1.279584×10^{-1}	9.683293×10^{-4}
Stored change ΔE (J)	4.284645×10^{-1}	2.089369×10^{-1}	7.197046×10^{-4}
Peak stored energy (J)	8.269431×10^{-1}	6.864552×10^{-1}	1.504024×10^{-2}
Mean input power (W)	5.780050×10^{-2}	3.546272×10^{-2}	1.993135×10^{-4}
Closure error (%)	1.634	4.991	15.299
Acceleration derivative correlation	0.998388	0.998162	0.987487
Model residual RMS / force RMS (%)	1.462	6.485	1.376

0.9875 — the lowest of the three by an order of magnitude in its complement — and the vertical energies are three orders of magnitude smaller than the lateral ones, at a peak stored energy of 1.50×10^{-2} J and a mean input power of 0.20 mW. Differencing a 1.4 μm record sampled at 1 kHz to obtain accelerations, and then integrating products of those accelerations over ten seconds, is a numerically hostile operation; the 15.3 % figure measures the differencing, not the identification.

This is the point. The same identified parameters that produce a 15.3 % closure error when evaluated against *differenced* derivatives produce a 0.248 % holdout error and a −0.081 % stiffness agreement when the derivatives come from the smooth surrogate. The audit therefore functions as a direct, quantitative demonstration of the argument advanced in Section 3.8 and Section 2.6: the classical derivative route degrades exactly where the signal is small and the sampling is coarse relative to the frequency content, and the automatic-differentiation route does not.

5.10 Modal Purity as the Explanatory Variable

The three directions do not perform identically, and the ordering of their performance is not accidental. Table 5.10 arranges the local modal diagnostics at the reference sensor alongside every downstream error measure.

Read down the Y column: every single diagnostic is the worst of the three, and by consistent margins. The cause is identified in the fourth row. At the reference sensor, the Mode-2 response accounts for a squared fraction of 0.6798 of the local motion, and the second contributor is Mode 3 at 30.516 Hz with a local amplitude of 2.380×10^{-3} — 59 % of the target mode's own amplitude. The X case has a far cleaner reference coordinate (squared fraction 0.8828, second contributor at only 26 % of the target amplitude), and the Z case has had its single significant contaminant explicitly removed by decomposition.

The vertical case illustrates the mechanism, with one caveat that must travel with it. Before decomposition, S_C07 is the *worst* of the three reference coordinates by local modal purity: the

Table 5.10: Local modal purity at S_C07 against downstream accuracy.

Quantity	**X**	**Y**	**Z (after decomposition)**
Target mode	1	2	4
Target squared fraction at S_C07	0.8828	**0.6798**	0.3540
Locally largest mode at S_C07	1 (rank 1)	2 (rank 1)	6 (target at rank 2)
Second contributor at S_C07	Mode 8, 1.069×10^{-3}	**Mode 3,** 2.380×10^{-3}	Mode 7, 3.440×10^{-3}
Modes explicitly separated	1	1	2
Fusion residual (%)	0.01924	**0.02863**	6.69×10^{-10}
Hidden-sensor error (%)	0.00974	**0.15588**	2.99×10^{-9}
LS relative force RMSE (%)	3.256	**7.981**	2.522
Model residual / force RMS (%)	1.462	**6.485**	1.376
Holdout relative RMSE (%)	0.2635	**0.3552**	0.2481
$\lvert\Delta K\rvert$ vs modal reference (%)	0.1755	**0.2071**	0.0812

Mode-4 squared fraction is 0.3540 and the locally dominant mode is Mode 6, not the target. A single-mode treatment of that raw record would have identified the wrong mode outright. After two-mode decomposition the spatial diagnostics become the strongest of the three — a 6.69×10^{-10} % reconstruction residual and a 2.99×10^{-9} % error at the withheld sensor. Those are the measures that vindicate the decomposition. Its final *stiffness* deviation does not, for the scaling reason retracted in Section 5.8, and the vertical point is plotted in Figure 5.11 with that caveat attached.

The conclusion, restricted to what the evidence supports, is that the fidelity of the reference-coordinate observation is governed by the modal purity of what preprocessing supplies, and that this purity is under the analyst's control rather than dictated by the structure. The X and Y stiffness deviations — 0.1755 % against a squared fraction of 0.8828, and 0.2071 % against 0.6798 — are the two points that support the further claim that purity tracks identification accuracy, and two points are a suggestive trend, not a demonstrated law. Y is marginally the weakest not because Mode 2 is harder to identify than Mode 1 or Mode 4, but because its reference coordinate retains Mode-3 content that was not explicitly separated — and the framework's response to that contamination is a 0.207 % stiffness error rather than a failure. That is a robustness result.

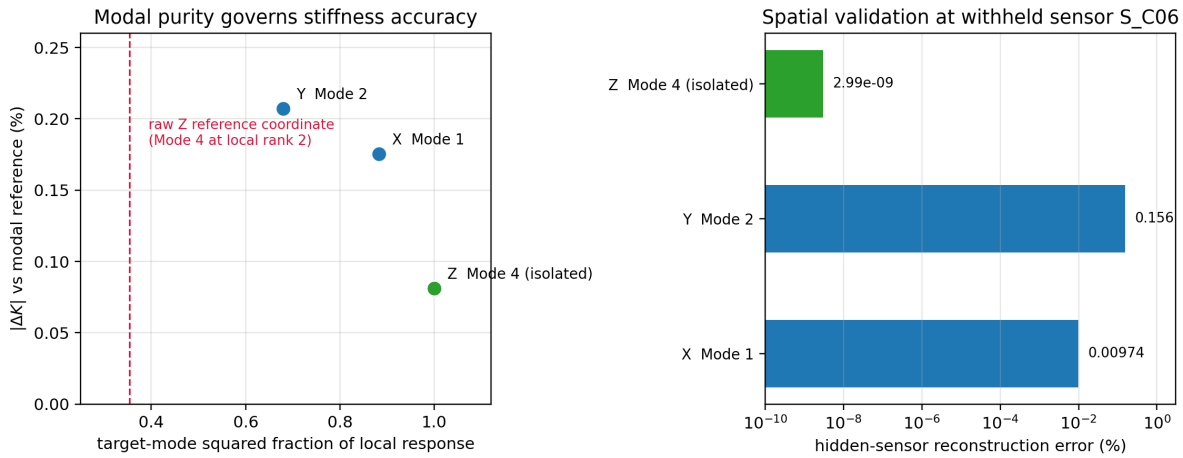


Figure 5.11: Modal purity at the reference coordinate against identification accuracy. Left: target-mode squared fraction of the local response against stiffness error, with the raw vertical reference coordinate marked. Right: reconstruction error at the withheld spatial sensor S_C06, on a logarithmic scale.

5.11 Evidence Against Memorisation

A displacement surrogate that reproduced its training data without engaging the physics term would leave the trainable parameters near their initial values, because the data term's gradient with respect to $\hat{k}$ and $\hat{c}$ is identically zero (Section 3.10). Six independent lines of evidence establish that this did not occur.

1. Parameter traverse. In X and Y the stiffness travelled factors of 8.760 and 9.336 from its initialisation to values agreeing with an independent modal reference to 0.18 % and 0.21 %. In all three directions the damping coefficient moved substantially — by factors of 0.331, 0.348 and 5.196.

2. Two unrelated estimators agree. The network and the unconstrained least-squares estimator, which share no derivatives, no objective and no solver, return stiffnesses agreeing to 0.041 %, 0.019 % and 0.009 %.

3. Agreement with a reference the network never saw. Neither K nor C nor f_n is supplied to training in any form. The modal reference stiffness enters the pipeline only as a post-hoc comparison in the evaluation stage.

4. Temporal generalisation. Errors at 2,000 withheld time samples exceed errors at observed samples by 0.07 %, 0.18 % and 2.4 % — ratios of 1.0007, 1.0018 and 1.0236. Memorisation would produce a far larger disparity.

5. Spatial generalisation. The modal coordinate recovered from four sensors predicts the motion at an entirely unseen fifth sensor to 0.0097 %, 0.1559 % and 3.0×10^{-9} %, a test that concerns the spatial hypothesis rather than the fit.

6. Displacement-only input. Velocity and acceleration are never supplied. The network cannot infer C from data it does not possess; the coefficient is determined solely through the residual orthogonality conditions (3.21).

5.12 Effective Damping and its Interpretation

The identified damping coefficients are 3,681.13, 3,894.64 and 4,433.64 $\mathrm{N\,s\,m^{-1}}$, corresponding to damping ratios of 0.0238 %, 0.0242 % and 0.0166 %. Three statements must accompany these numbers.

They are effective reduced-model quantities. The source transient analysis carried zero physical damping and a numerical damping factor of 0.005. The dissipation present in the records is therefore algorithmic. The identified C is the coefficient that reproduces that dissipation within the reduced model, and is not claimed as a material or structural damping property of steel.

They are physically admissible, which the classical route was not. The softplus reparameterisation guarantees positivity by construction, and all three identified values are positive and of consistent magnitude. The unconstrained least-squares estimator applied to the same data returned damping ratios of approximately -2.3 % in every direction.

They are substantially less well determined than the stiffnesses, and in the vertical direction the damping coefficient is not converged at all. Over the final tenth of iterations the stiffness varied by 0.041 %, 0.017 % and 0.009 %. Over the same window the damping varied by 1.17 %, 0.82 % and **16.95** %. Narrowing to the final twentieth does not resolve the vertical case either: its damping still spans 9.04 %, an order of magnitude worse than the 0.66 % and 0.68 % of the lateral runs.

The vertical figure deserves to be stated as a finding rather than a footnote. A coefficient that moves by 17 % across the last 2,800 iterations of a run whose stiffness has settled to four decimal places is not a converged parameter; it is a parameter the record cannot pin down. Reporting $C_Z = 4{,}433.64\,\mathrm{N\,s\,m^{-1}}$ to six figures alongside that diagnostic would be misleading, and the value should be read as an order-of-magnitude estimate, bracketed by roughly ± 9 % at the stopping point, not as a determination.

The cause is signal-to-noise, and it is specific to the vertical axis. Three quantities in this chapter converge on the same explanation. The isolated Mode-4 reference record has a root-mean-square amplitude of 1.436 μm and a peak of 2.644 μm — a factor of 13.9 smaller in RMS than the X record, and the smallest signal in the study by an order of magnitude in energy terms. Its acceleration derivative correlation against the solver export is 0.9875, the poorest of the three and the only one below 0.998. And its total dissipated energy over the full ten-second record is 9.68×10^{-4} J at a mean input power of 0.20 mW, three orders of magnitude below the lateral cases. Damping is identified from dissipated energy; when the dissipated energy is sub-millijoule and the displacement is at the micron scale, the residual term carrying $C\dot{u}$ is a small perturbation on a small signal, and the optimiser has correspondingly little gradient information to act on. The 16.95 % fluctuation is that scarcity made visible.

The ordering across directions is otherwise exactly what the observability analysis of Section 3.7 predicts in advance: stiffness accumulates evidence per oscillation cycle, of which the records contain 201, 206 and 514, while damping accumulates evidence per decay time constant, of which the records contain only 0.30, 0.31 and 0.54. The vertical case observes the most cycles and has the largest ratio T/τ, which is why its stiffness is the most tightly bounded; it also

carries the smallest absolute dissipated energy, which is why its damping is the least.

Reporting both parameters with their convergence diagnostics, rather than reporting a single number for each, is the appropriate presentation for structural assessment: the diagnostics state how much of each parameter the measurement actually supports.

5.13 Limitations

The following limitations are recorded.

1. One reduced relation per direction. The framework identifies a single-degree-of-freedom relation per excitation direction. Modes not retained are removed by preprocessing rather than identified. Mode 7 in the vertical case is a preprocessing quantity throughout and is never an output of the network.

2. Damping is algorithmic, not material. As set out above, the identified C reflects the numerical dissipation of the source solution. Identification of material damping requires a source model carrying physical damping.

3. Single seed. All three production runs used seed 0. A multi-seed ensemble would quantify initialisation sensitivity and supply empirical uncertainty bounds; the present results establish accuracy against independent references but not initialisation variance.

4. The vertical stiffness scale begins near its target. As Section 5.8 records, the vertical non-dimensionalisation places $\hat{k}_{\mathrm{ref}}$ at 1.0095 against an initialisation of 1.0. The vertical stiffness result is therefore validated by its agreement with independent references and by its convergence stability, not by the length of its traverse.

5. Exported derivatives are discrete solver quantities. The velocity and acceleration columns used by the diagnostic and by the energy audit are time-integration outputs, not continuous derivatives. This is the acknowledged reason for the diagnostic's negative damping and for the vertical energy-closure error, and it does not affect the network, which never consumes them.

6. Noise-free data. All records are numerically exact finite element output. Field measurements carry sensor noise, quantisation and drift, and a systematic noise study is a prerequisite for deployment on measured data.

7. Coarse force knots. The original force table used approximately 0.01 s knots against a 0.001 s storage grid, so the extracted force history carries the solver's tabular interpolation. This was retained deliberately rather than idealised.

5.14 Synthesis

The results support a single, structured conclusion.

A physics-informed neural network trained on displacement alone recovered the reference-coordinate stiffness of a three-dimensional finite element structure to within 0.081 %–0.207 %

of an independently computed modal reference in all three spatial directions, and to within 0.009 %–0.041 % of an independently computed least-squares estimate, while generalising to withheld time samples at coefficients of determination of 0.99999 in every case. The same network, loss, optimiser and code path were used throughout.

Three qualifications bound what that sentence may be taken to mean.

The stiffness result is not the network's achievement to claim alone. The reduction to a well-posed scalar problem was performed classically before training began, and ordinary least squares solves the residual two-parameter problem to the same accuracy in 0.2 ms against 223–367 s (Table 5.7). On stiffness the network reproduces a result that is already available six orders of magnitude more cheaply, and Section 1.4 sets out why it is nonetheless worth computing.

The vertical figure is partly a scaling artefact. Section 5.8 retracts any reading of the −0.081 % vertical deviation as evidence of decomposition quality: that run began 0.94 % from its target because $\hat{k}_{\mathrm{ref}} = 1.0095$ coincides with $\hat{k}_0 = 1.0$. The decomposition is vindicated by its spatial diagnostics and by the independent Q_7/Q_4 cross-check, not by the stiffness number.

Vertical damping did not converge. It fluctuated by 16.95 % across the final tenth of iterations, and by 9.04 % across the final twentieth, driven by a micron-scale displacement and a sub-millijoule dissipated energy budget (Section 5.12).

What the evidence does support is a narrower and more durable claim. Coordinate consistency, constructed before learning, is what makes the reduced inverse problem well posed in each direction, and two pieces of evidence establish that attribution. The vertical load-projection ratio of 0.7685 shows that a formulation neglecting generalised-force projection would have mis-identified the vertical stiffness by approximately 30 % while fitting the displacement record equally well — an error invisible in every fit statistic this chapter reports. And the spatial reconstruction errors at the withheld sensor, spanning 0.0097 % to 3.0×10^{-9} %, track the modal content that preprocessing removed rather than any property of the network.

The damping results complete the picture in a way a uniformly tight outcome would not have. Damping was recovered as a positive, physically admissible quantity in all three directions — which the classical estimator could not manage at all — with convergence diagnostics spanning a factor of twenty across the three cases, in accordance with the observability scalings derived in advance from the governing equation. Reporting that spread, rather than three tidy numbers, is the point: a framework that states how much of each parameter the measurement actually supports is more useful for structural assessment than one that returns a confident number regardless.

Chapter 6

Conclusion and Future Work

6.1 Summary of Contributions

This chapter set out to establish that reduced-order inverse identification of a three-dimensional structure can be made accurate and uniform across spatial directions by constructing coordinate consistency before learning, rather than assuming it. The following were delivered.

A coordinate-consistent reduction. Observation, equivalent mass, equivalent stiffness and equivalent force were all referred to one physical coordinate — sensor S_C07 at node 3461 — for all three directions, through the relations $m^* = \phi_{\text{ref}}^{-2}$, $k^* = m^*\omega_r^2$ and $F_{\text{eq}} = Q_r/\phi_{\text{ref}}$. Each identity was verified numerically against the stored modal quantities to machine precision.

Exact finite element load-vector projection. The equivalent force was obtained by projecting the assembled unit-pressure load vector over 180 loaded elements onto each retained mode under the solver's own node ordering, yielding projection ratios of 1.0016, 1.0086 and 0.7685. The vertical ratio demonstrates that a force heuristic would have produced a 30 % stiffness error undetectable in any fit statistic.

Multi-sensor modal fusion and two-mode separation as preprocessing. Four sensors were fused into a single modal coordinate, and in the vertical direction an overdetermined four-sensor system with condition number 1.534 was solved for two modal coordinates, of which only the Mode-4 component was retained. The network remained a scalar one-input, one-output map throughout; no architectural complexity was added.

Identification across three axes, with its limits stated. Stiffness agreed with the modal reference to $-0.175\,\%$, $-0.207\,\%$ and $-0.081\,\%$; frequency to $-0.088\,\%$, $-0.104\,\%$ and $-0.041\,\%$; and with an independent least-squares estimate to $+0.041\,\%$, $+0.019\,\%$ and $-0.009\,\%$. Holdout coefficients of determination were 0.999993, 0.999987 and 0.999994 at relative errors of 0.264 %, 0.355 % and 0.248 %. Two limits are recorded alongside these figures rather than after them: the vertical stiffness deviation is inflated in significance by a non-dimensionalisation that began that run 0.94 % from its target, and the vertical damping coefficient did not converge, fluctuating by 16.95 % over the final tenth of iterations.

An explicit accounting of what the network contributes over the classical alternative. Ordinary least squares solves the same reduced two-parameter problem in 0.18–0.22 ms against

223–367 s, a cost ratio of order 10^6, and returns the same stiffness to within 0.009 %–0.041 %. The network's contribution is confined to one property: it never differentiates the sampled record, and therefore returns a positive, physically admissible damping coefficient where the derivative-consuming estimator returns -2.3 % of critical in all three directions. That is the whole of the case, and it is made without inflation.

Dual, non-overlapping validation. Spatial generalisation was demonstrated at a physically distinct withheld sensor to 0.0097 %, 0.1559 % and 3.0×10^{-9} %; temporal generalisation at 2,000 withheld samples with holdout-to-training error ratios of 1.0007, 1.0018 and 1.0236.

Independent cross-verification of the modal decomposition. The vertical two-mode separation was confirmed by comparing a measured modal-coordinate ratio of 4.041 % against a value of 4.019 % predicted from the projected modal forces and analytical dynamic compliance — routes sharing no intermediate quantity — at a relative discrepancy of 0.55 %.

An explanatory account of the residual error. Identification accuracy across the three directions was shown to track the modal purity of the observation supplied to the network, a quantity controlled entirely by preprocessing, rather than any property of the network or of the mode being identified.

6.2 Future Work

Simultaneous multi-mode identification. The present framework removes non-target modes by preprocessing. A natural extension identifies several modal coordinates jointly, retaining one network output per retained mode with a coupled residual. The vertical direction, with its cleanly separated Mode-4 and Mode-7 content and its well-conditioned modal matrix, is the natural first test case, and would allow the Mode-7 parameters to become outputs rather than preprocessing quantities.

Material damping identification. The identified C presently reflects algorithmic dissipation. Re-solving the transient analyses with a prescribed Rayleigh damping model would allow the identification to be tested against a known material damping target. Identification of the Rayleigh coefficients α and β separately requires at least two modes, since a single mode yields one equation in two unknowns — a further argument for the multi-mode extension above.

Noise robustness. All present records are numerically exact. A systematic study sweeping additive noise, quantisation and sampling rate would establish the signal-to-noise ratio at which each of the four consistency requirements of Section 1.2 begins to degrade, and is a prerequisite for deployment on measured data. Given the results of Section 5.9, the expectation is that the automatic-differentiation route degrades substantially more gracefully than the classical alternative.

Multi-seed ensembles. Repeating each direction across an ensemble of initialisations would convert the present point estimates into empirical distributions and supply uncertainty bounds on both parameters, with the damping coefficient the more informative case given the convergence ranges reported in Table 5.8.

Optimal reference-coordinate selection. The reference coordinate was chosen here by engineering judgment under explicit placement rules. The modal purity analysis of Section 5.10 suggests that the target-mode squared fraction at a candidate node is a strong predictor of downstream identification accuracy, and could serve as an objective for formal reference-coordinate selection along the lines of the effective-independence and information-entropy criteria reviewed in Section 2.4.

Experimental validation. The natural culmination is application to a physically instrumented structure, where the modal amplitudes entering every relation in Section 3.3 must themselves be identified from measurement rather than read from a solver.

6.3 Closing Remarks

The finding of this chapter is narrower and more useful than "the method works". The same network, loss, optimiser and configuration path sufficed in three directions whose raw records differ substantially — by a factor of 2.56 in frequency, a factor of 13.9 in root-mean-square displacement amplitude, and a factor of 2.49 in local modal purity at the same reference node. That sufficiency was not bought with network capacity, extra iterations or direction-specific tuning. It was bought by ensuring that what the network was asked to identify was, in each case, a well-posed scalar problem in a single consistent coordinate — and the work of establishing that coordinate was classical, not learned.

It is worth being exact about the division. ANSYS solved the elastodynamics. Block Lanczos supplied the modal basis, and with it the reference stiffness that the network would later be compared against. Least squares fused four sensors into one modal coordinate, and an exact load projection supplied the forcing. What the network received was a single-degree-of-freedom equation with two unknown scalars — a problem ordinary least squares dispatches in a fifth of a millisecond. The network's justification is not that it solved something harder. It is that it solved the same thing without ever differentiating the data, and so returned a damping coefficient that was physically admissible where the classical route returned one that was not.

The vertical direction is the case that most needs care, and an earlier reading of these results got it wrong. By local modal purity it was the worst reference coordinate of the three, with the target mode ranked second in local amplitude behind a mode that does not participate in the excitation at all; treated naively it would have identified the wrong mode outright. Two-mode separation fixed that, and the spatial diagnostics prove it: a $2.99 \times 10^{-9}\,\%$ reconstruction error at a withheld sensor, and a $0.55\,\%$ agreement between two routes that share no intermediate quantity. But its headline stiffness figure of $-0.081\,\%$ does *not* prove it, because the vertical non-dimensionalisation placed the initialisation $0.94\,\%$ from the answer; and its damping coefficient, fluctuating by $16.95\,\%$ at the stopping point, is not a determination at all. Both are stated here as they are stated in Chapter 5: as limits of the study, not as results of it.

For structural assessment the implication is practical. When a reduced-order identification disagrees with expectation, the first question is not whether the estimator has failed but whether the reduction was consistent — and that question is answerable in advance, from the modal

model and the load distribution, before any data is fitted. The construction developed here makes that check a routine part of the pipeline rather than a post-hoc investigation.

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